

Session 2 — Judging a model

From the first success metric to the right amount of complexity. Videos V17 to V29.

Thursday 10 September 2026 · student copy

Questions during the session. The live interpretation adds a delay, so please send your questions in the Zoom chat, or better, after the session on the WhatsApp number you already have. Every question gets an answer.

How to use this copy: each block gives you one idea, a story, three things to keep, three questions and one thing to try. The answers are not printed here — we work through them together in the session. The pages on the right of each slide are from the video decks you watched; the speaker notes under every slide explain them in plain words and link short videos for each term.

What we do today

One block per video. Each block: the one idea, a story, three things to keep, three questions you answer, and one thing to try.

- | | |
|--|---|
| 1 Success metric V17 · 15 min | 8 Overfitting V24 · 15 min |
| 2 Confusion matrix V18 · 15 min | 9 Underfitting V25 · 12 min |
| 3 Accuracy trap V19 · 15 min | 10 Train-test split V26 · 12 min |
| 4 Precision vs recall V20 · 15 min | 11 Validation curve V27 · 12 min |
| 5 F1 score V21 · 12 min | 12 Bias and variance V28 · 12 min |
| 6 Threshold and money V22 · 15 min | 13 Choosing complexity V29 · 12 min |
| 7 Talking to stakeholders V23 · 12 min | |

The blocks build on each other: what to measure, how to read the four boxes, what accuracy hides, where to set the line, what it costs, how to say it, and how much model the problem can carry.

Block 1 of 13, 15 minutes. Video V17.

Choosing a success metric

Before you build anything: what does winning mean?



the deck you watched: V17, Choosing a Success Metric

The one idea

Decide what "winning" means before you build the model. Write it down.

The business says

"Winning is fewer stolen euros" or "fewer unpaid loans" or "fewer very late deliveries".

The data team says

"Then this is the number we will report, and this is the bar it must clear."

Both sign. That piece of paper is the metric. Everything else in this block is about getting it right.

FROM VIDEO V17

The Metric Is a Contract

An agreement between data science and business. Signed before any data is touched.



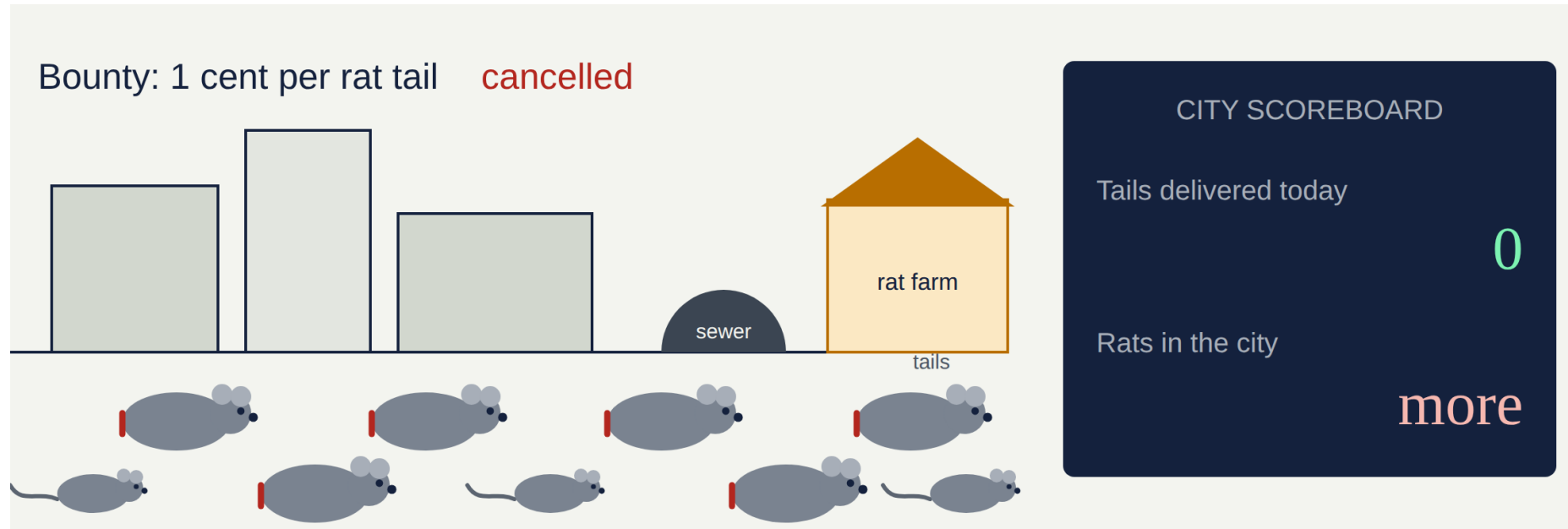
V17, page 4 — The metric is a contract

Don't let that happen.

Meet with business owners **before any model training**. Agree on the metric, the rationale, and the acceptable performance bar. Write it down. Get sign-off.

V17, page 16 — Don't let that happen

A story: Hanoi, 1902



- 1 Hanoi, 1902. The French governor has just built modern sewers. The rats love them, plague is in the region, and the city is overrun.
- 2 The city sends hired rat catchers into the sewers. On the best day they kill more than twenty thousand rats. The rats keep coming.
- 3 So the city opens the hunt to everyone, with a bounty: one cent per rat tail handed in. Tails pour in by the thousand. On the scoreboard, success.
- 4 Then people notice rats running around with no tails. The catchers cut off the tail and let the rat go, so it can breed more rats, and more tails.
- 5 Inspectors find rat farms on the edge of the city. The bounty is cancelled. Lesson: they paid for tails. They wanted fewer rats.

Same mistake, in a company

"Our model is 95% accurate." Champagne. Six months later, the company is losing money.

What they measured

95% of predictions correct. The rat tails.

What the business wanted

Less money lost. Fewer rats.

The model was accurate at the wrong thing. Nobody had agreed what winning meant.

FROM VIDEO V17

95% Accuracy. Champagne. Then... Losses.

High technical accuracy $\neq$ solving the business problem.

⚠ The model was accurate at the **wrong thing**.



V17, page 2 — 95% accuracy. Champagne. Then losses.

Clarity Before Code



Ask the hard questions early

- Catch every fraud case, even with false alarms?
- Approve customers fast, even if bad actors slip through?

That question belongs in the boardroom — not in the code.

V17, page 5 — Clarity before code

Which mistake costs more?

A false alarm

The model said yes. It was no.

A miss

The model said no. It was yes.

False alarms cost more

Measure: of everything we flagged, how much was right?

precision

Misses cost more

Measure: of all the real cases, how many did we catch?

recall

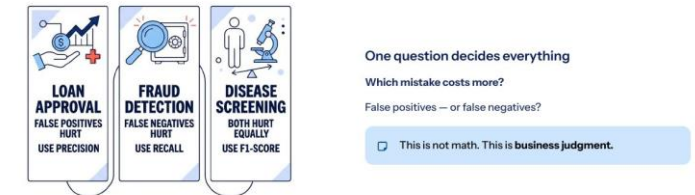
Both hurt

Measure both and balance them.

F1

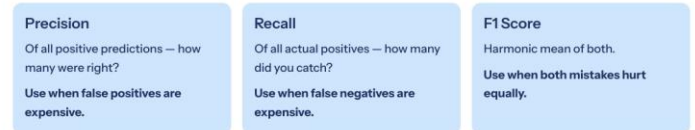
FROM VIDEO V17

Real-World Metric Map



V17, page 11 — Which mistake costs more?

Precision vs. Recall



V17, page 10 — Precision vs recall vs F1

If you remember nothing else

Three things to keep

- 1 Decide what winning means before building. Write it down, both sides sign.
- 2 Ask "which mistake costs more?" The answer picks the metric. That is a business call, not a maths call.
- 3 Never choose the metric after seeing the results.

FROM VIDEO V17

Key Takeaways

1	2
The metric is a contract Agreed before any code is written.	Match metric to mistake cost Precision, Recall, F1, MAE, or RMSE — each has a purpose.
3	4
Cherry-picking is self-deception Commit to one metric. Before you train. No exceptions.	Get sign-off Document it. No surprises. No arguments.

V17, page 23 — Key takeaways

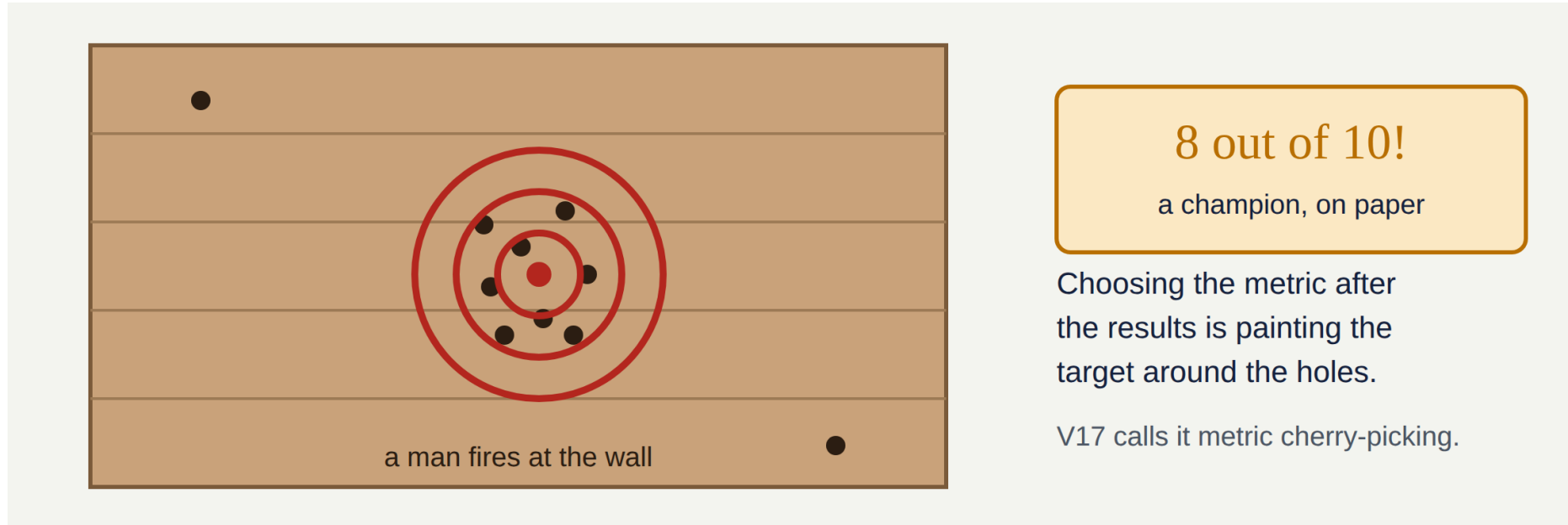
The Metric Is a Contract

An agreement between data science and business. Signed before any data is touched.



V17, page 4 — The metric is a contract

Why rule 3 matters: the Texas sharpshooter



- 1 A man fires ten shots at the side of a barn. No target. The holes land wherever they land.
- 2 Then he walks up with a pot of red paint and paints a target around the tightest group of holes.
- 3 Eight out of ten. He calls himself a sharpshooter. A team that tries ten metrics and reports the flattering one is doing exactly this.

Question 1 of 3

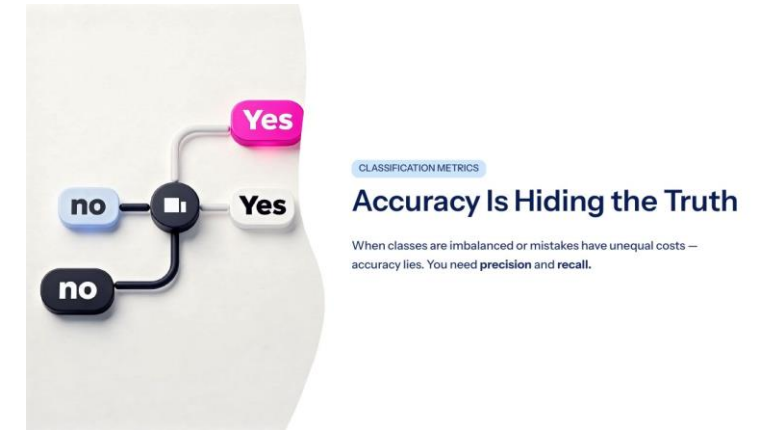
The data team reports: "Our model is 95% accurate." What is the first question to ask?

- A** Which algorithm did you use?
- B** What did we agree winning means, and does 95% deliver it?
- C** Can you get it to 99%?
- D** How long did the training take?

Hint. Think of the rat tails. A big number is only good news if it counts the right thing.

Answer your letter in the Zoom chat. We work through the answer together in the session.

FROM VIDEO V17



V17, page 9 — Accuracy is hiding the truth

Question 2 of 3

A bank lets a model say "approve" or "reject" for small loans. Which mistake costs the bank more?

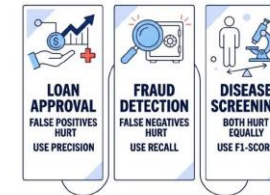
- A** Approving someone who then never pays back
- B** Rejecting someone who would have paid back
- C** They cost exactly the same
- D** Neither. Only the speed of the decision matters

Hint. What does the bank actually lose in each case? The whole loan, or some interest?

Answer your letter in the Zoom chat. We work through the answer together in the session.

FROM VIDEO V17

Real-World Metric Map



One question decides everything

Which mistake costs more?

False positives — or false negatives?

☐ This is not math. This is business judgment.

V17, page 11 — Which mistake costs more?

Question 3 of 3

A delivery company predicts arrival times. Its model is late by 150 minutes in total this week. Which version of that is worse for the business?

- A** Ten deliveries, each 15 minutes late
- B** One delivery, two and a half hours late
- C** Equally bad: 150 minutes late either way
- D** Neither matters as long as the average is fine

Hint. Which customer stops using the service?

Answer your letter in the Zoom chat. We work through the answer together in the session.

FROM VIDEO V17

REGRESSION METRICS

Predicting Numbers? Know Your Error.

MAE

Mean Absolute Error

All errors treated equally

RMSE

Root Mean Squared Error

Large errors punished heavily

V17, page 7 — Regression metrics: MAE vs RMSE

Your turn: write the contract

Pick a business problem

Fraud on card payments ▾

Watch the paper change. Then do the same for a problem from your own job: three lines, in words, no formulas.

Success metric

We are predicting

The expensive mistake

So we measure

Winning means

whether a card payment is fraud

a miss: a real fraud goes through

of all real frauds, how many we caught (recall)

we catch at least 9 of every 10 frauds, with false alarms on fewer than 1 payment in 100

agreed before building

business owner

data team

Before we move on

Three things from this block

- 1 Paying for rat tails is not the same as having fewer rats. Measure the thing you want.
- 2 "Which mistake costs more?" is the question that chooses the metric. Managers answer it, not programmers.
- 3 Pick the metric before the results, or you are painting targets around bullet holes.

FROM VIDEO V17

Key Takeaways

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The metric is a contract Agreed before any code is written.	Match metric to mistake cost Precision, Recall, F1, MAE, or RMSE — each has a purpose.
3	4
Cherry-picking is self-deception Commit to one metric. Before you train. No exceptions.	Get sign-off Document it. No surprises. No arguments.

V17, page 23 — Key takeaways



The Metric Is Your Compass

Once locked in, every decision follows:

- Model Architecture**
Random forest, neural net, or logistic regression?
- Evaluation Strategy**
Cross-validation, hold-out test set, or both?
- Production Monitoring**
Track the metric in the real world after deployment.

V17, page 22 — The metric is your compass

Block 2 of 13, 15 minutes. Video V18.

Reading a confusion matrix

Four boxes that show what one number hides.



the deck you watched: V18, Reading a Confusion Matrix

The one idea

"95% accurate" tells you how often the model was right. It cannot tell you which way it was wrong.

Too eager?

Your spam filter blocks the email from your boss.

Too quiet?

Your spam filter lets the phishing attack through.

Both filters can be 95% accurate. The confusion matrix is the table that tells them apart.

FROM VIDEO V18

Accuracy Hides the Story

What accuracy shows
Total correct predictions

What accuracy hides
The *direction* of your mistakes



V18, page 2 — Accuracy hides the story

One Number. Two Hidden Failures.

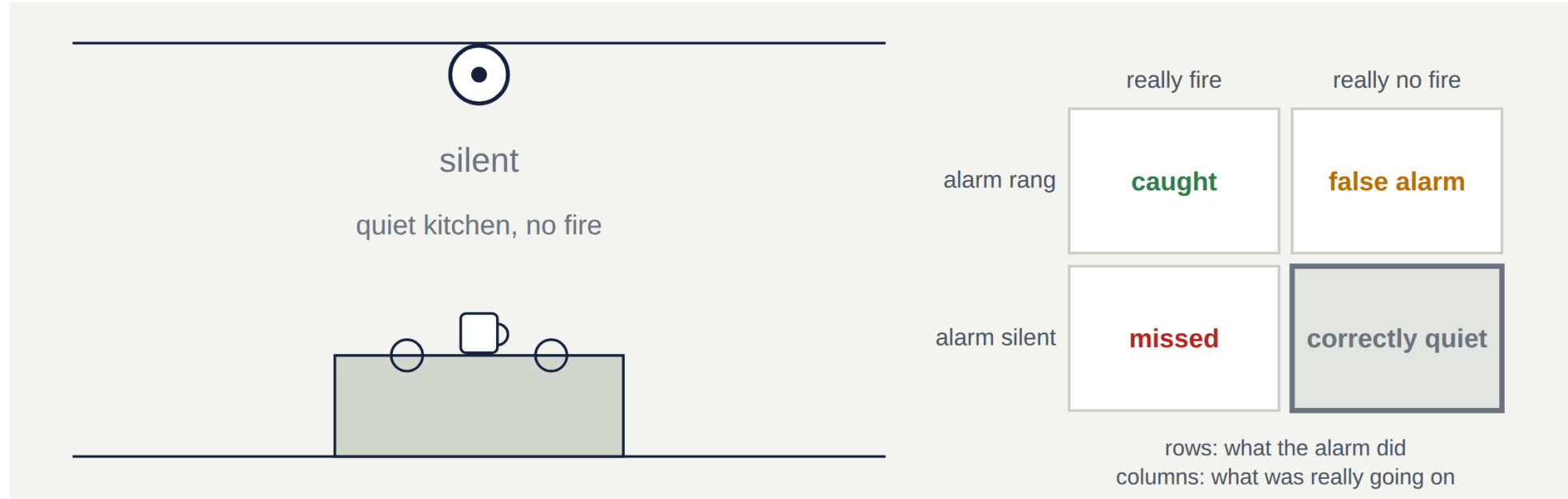


The real questions

- Is your filter **over-flagging** legitimate emails?
- Is it **missing** real phishing attacks?
- A single accuracy score can't answer either.

V18, page 3 — One number, two hidden failures

The four boxes, with a smoke alarm



- 1 A kitchen, a smoke alarm, and a table with four boxes. Rows: what the alarm did. Columns: what was really going on.
- 2 Real fire, alarm rings. The alarm got it right. Top-left box: caught.
- 3 Burnt toast, alarm rings. There was no fire. Top-right box: a false alarm. Annoying, not dangerous.
- 4 Real fire, alarm silent. The danger was there and nothing happened. Bottom-left box: a miss. This is the dangerous one.
- 5 Quiet kitchen, alarm silent. Nothing to flag, nothing flagged. Bottom-right box: correctly quiet. Two boxes are wins, two are mistakes, and the two mistakes point in different directions.

A story you know: the boy who cried wolf



- 1 A shepherd boy watches the village sheep on the hill. Nothing happens for days. He is bored.
- 2 He shouts "Wolf!" The villagers drop their work and run up the hill. No wolf. A false alarm: nobody hurt, an afternoon wasted.
- 3 A week later he does it again. Same result. Now the villagers have learned something: this alarm is noise.
- 4 Then a real wolf comes. The boy shouts louder than ever. Nobody comes. The false alarms did not only waste time; they turned the next real alarm into a miss.

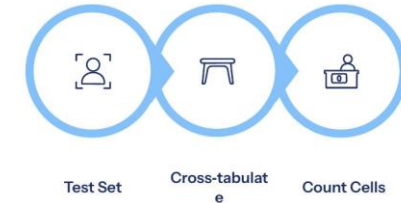
If you remember nothing else

Three things to keep

- 1 Rows are what the model said. Columns are what was true. Read the two "wrong" boxes first.
- 2 A false alarm wastes effort. A miss does harm. They are not the same mistake.
- 3 Accuracy adds the two "right" boxes and hides the two "wrong" ones. Always ask to see the four boxes.

FROM VIDEO V18

How It Works



Rows represent what the model **predicted**. Columns represent what was **actually true**. Each cell is a count.

V18, page 5 — How it works

Four Cells → Four Insights



V18, page 16 — Four cells, four insights

Question 1 of 3

The fraud model from the video is tested on 1,000 payments: 80 frauds caught, 20 good payments wrongly blocked, 10 frauds missed, 890 good payments correctly cleared. What is its accuracy?

- A 80%
- B 89%
- C 97%
- D 99%

Hint. Accuracy = the two "right" boxes added together, out of everything.

Answer your letter in the Zoom chat. We work through the answer together in the session.

FROM VIDEO V18

Real Numbers: Fraud Detection

A model tested on 1,000 transactions:

80

True Positives
Real fraud caught ✓

20

False Positives
Good customers flagged ⚠️

10

False Negatives
Fraud missed ❌

890

True Negatives
Legit transactions cleared ✓



V18, page 12 — Real numbers: fraud detection

Question 2 of 3

Which of these is a miss (the alarm stayed silent during a real fire)?

- A** The spam filter blocks a real email from your boss
- B** A screening test tells a sick patient they are healthy
- C** The fraud model blocks a payment that really was fraud
- D** The loan model approves a customer who pays back on time

Hint. A miss is "model said no, truth was yes".

Answer your letter in the Zoom chat. We work through the answer together in the session.

FROM VIDEO V18

The 2×2 Table

	Actually Positive	Actually Negative
Predicted Positive	True Positive ✓	False Positive ⚠
Predicted Negative	False Negative 🚫	True Negative ✓

Four cells. Four outcomes. The full picture of model performance.

V18, page 6 — The 2×2 table

Question 3 of 3

Two businesses use a model with the same four boxes. For which one is a miss (silent alarm) the worse mistake?

- A** A spam filter for a company inbox
- B** A screening test for a serious disease
- C** Both, exactly the same
- D** It depends on which algorithm was used

Hint. What happens to a person after each kind of mistake?

Answer your letter in the Zoom chat. We work through the answer together in the session.

FROM VIDEO V18

BUSINESS REALITY

Not All Errors Are Equal

False Positive = Wasted Effort

Spam filter blocks your boss's email. Loan officer reviews an obviously safe application. Users get frustrated.

False Negative = Real Harm

Fraud slips through. A disease goes undetected. Criminals bypass security. Consequences, not inconvenience.

V18, page 11 — Not all errors are equal

Your turn: what does accuracy hide?

1,000 card payments, 90 of them fraud. The model's job is to flag fraud. You decide how many of the 90 frauds it misses and how many honest customers it wrongly blocks; the four boxes and the accuracy figure update as you move the sliders.

10 frauds the model misses, out of 90



20 good customers wrongly blocked, out of 910



97%

accurate. 10 frauds through, 20 customers annoyed.

	Really fraud	Really honest
Model said fraud	80 caught	20 false alarm
Model said honest	10 missed	890 correctly cleared

Try, in this order: (1) misses to 90: the model catches no fraud at all, what does accuracy say? (2) misses to 0 and false alarms to 200: every fraud caught, 200 customers angry, accuracy? (3) find a setting with 97% accuracy and 40 frauds missed. The point: accuracy barely moves while the damage moves a lot.

Before we move on

Three things from this block

- 1 One number hides. Four boxes show. Ask for the four boxes.
- 2 A false alarm is a wasted trip up the hill. A miss is the wolf among the sheep.
- 3 Which mistake is worse depends on the business, never on the model.

FROM VIDEO V18

Key Takeaway

**One number
hides. Four cells
reveal.**

The confusion matrix turns an opaque accuracy score into **actionable intelligence** — showing you exactly where your model wins and where it fails.

V18, page 17 — Key takeaway

Four Cells → Four Insights



V18, page 16 — Four cells, four insights

Block 3 of 13, 15 minutes. Video V19.

Why accuracy misleads on rare events

When the thing you hunt is rare, "accurate" can mean "never tried".



the deck you watched: V19, Why accuracy misleads on imbalanced data

The one idea

When the thing you hunt is rare, "accurate" can mean "never tried".

The lazy model

Says "honest" to all 1,000 payments. Right 990 times. 99% accurate.

The result

Zero frauds caught. Zero customers protected. Champagne anyway.

Fraud is under 1% of payments. Churn is 5 to 10%. Defaults under 5%. Rare is the normal case in business, so this trap is the normal case too.

FROM VIDEO V19



The Fraud Model That Failed Perfectly

The Setup

Bank deploys fraud detection model. Test accuracy: **99%**

The Reality

Model flags **zero** transactions as fraud. Not a single case caught.

The Paradox

Metric says **success**. Business says **failure**.

V19, page 2 — The fraud model that failed perfectly

Why 99% Accuracy Is Meaningless Here



The Math Is Simple

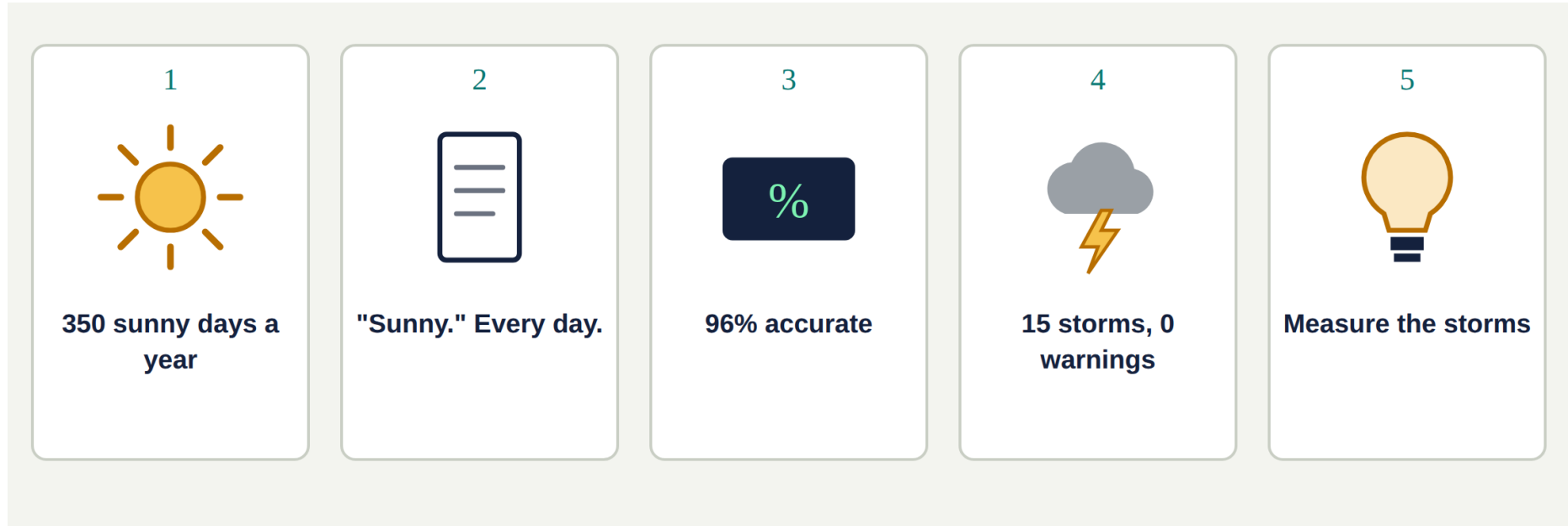
Only **1%** of transactions are genuinely fraudulent.

Predict "**safe**" on **everything** → correct 99% of the time.

The model learned **nothing**. It just bet on the majority.

V19, page 3 — Why 99% accuracy is meaningless here

A story: the forecaster who never looked outside



- 1 A desert town with 350 sunny days a year hires a weather forecaster.
- 2 He prints the same forecast every morning: sunny. He never looks out of the window.
- 3 At the end of the year the town checks: he was right 350 days out of 365. 96% accurate. A bonus is proposed.
- 4 Then someone counts the storms. Fifteen. He warned of none of them. The only days that mattered were the days he got wrong.
- 5 Lesson: when the important events are rare, being right most of the time is easy and worthless. Count the storms you caught.

If you remember nothing else

Three things to keep

- 1 Accuracy is the two right boxes over everything. When one answer is very common, the common answer does all the scoring.
- 2 A model that never says "yes" scores high on rare events and catches nothing.
- 3 Ask two questions instead: of all the real cases, how many did we catch? Of all our flags, how many were right?

FROM VIDEO V19



What Accuracy Hides

Missed Fraud Cases

How many real frauds slipped through undetected?

False Alarms Generated

How many innocent customers were wrongly flagged?

The Illusion

Accuracy lumps everything together and calls it success.

V19, page 7 — What accuracy hides

CORE LESSON

Stop trusting accuracy as a shortcut.

→ Always inspect the confusion matrix

→ Always report precision, recall, or F1

→ Match your metric to what the business can't afford to miss

V19, page 17 — Core lesson

Question 1 of 3

One customer in fifty cancels each month. A churn model predicts "stays" for every customer. What is its accuracy?

- A** 2%
- B** 50%
- C** 98%
- D** It cannot be computed without more data

Hint. How many customers out of 50 does the "stays" answer get right?

Answer your letter in the Zoom chat. We work through the answer together in the session.

FROM VIDEO V19

How Accuracy Is Calculated

The Formula

$$\text{Accuracy} = \frac{\text{Correct Predictions}}{\text{Total Predictions}}$$

Treats every prediction **equally** — rare or common.

When It Breaks Down

- Balanced classes → works fine
- One class dominates → becomes a **trap**
- Rewards predicting the majority — always

V19, page 5 — How accuracy is calculated

Question 2 of 3

Which of these businesses can safely judge a model by accuracy alone?

- A** A bank flagging fraud in 1% of payments
- B** A hospital screening for a disease found in 2% of patients
- C** A retailer predicting which of two equally popular colours a customer picks
- D** An insurer predicting the 3% of claims that are fraudulent

Hint. Accuracy works when yes and no are roughly equally common.

Answer your letter in the Zoom chat. We work through the answer together in the session.

FROM VIDEO V19

Imbalanced Data Is Everywhere



V19, page 6 — Imbalanced data is everywhere

Question 3 of 3

Two fraud models. Model A: 99% accurate, catches 0 of 100 frauds. Model B: 90% accurate, catches 90 of 100 frauds but raises 90 false alarms. Which would the fraud team rather have?

- A** Model A, it is more accurate
- B** Model B, it actually catches fraud
- C** Neither, both are useless
- D** It depends only on which algorithm they used

Hint. What is the fraud team paid to do?

FROM VIDEO V19

A Real Failure in Numbers

Model A

Accuracy: 99%

Recall on fraud: 0%

Worthless for fraud teams.

Model B

Recall: 90%

Precision: 40%

Catches fraud – but drowns in false alarms.

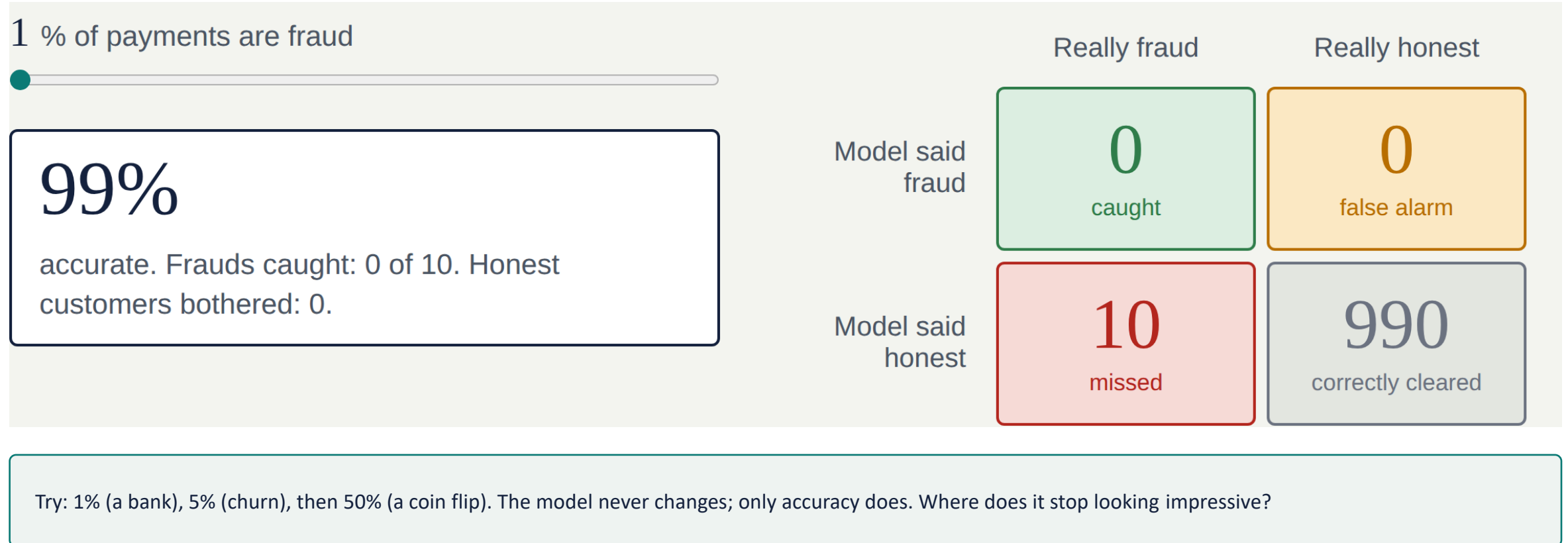
☐ One metric hides both failures. You need the full picture.

V19, page 13 — A real failure in numbers

Answer your letter in the Zoom chat. We work through the answer together in the session.

Your turn: the model that never says yes

1,000 payments. The model answers "honest" to every one of them. Slide the share of payments that are really fraud and watch what accuracy says about a model that catches nothing.



Before we move on

Three things from this block

- 1 On rare events, accuracy rewards a model for doing nothing.
- 2 Count the storms: of all the real cases, how many did we catch?
- 3 The right measure depends on what the business cannot afford to miss.

FROM VIDEO V19

The Bottom Line

Accuracy is intuitive. On imbalanced data, it's useless. The metric you choose is a business decision — make it deliberately.

Define cost first

Use precision + recall

Visualize with confusion matrix

Tune last, not first



V19, page 20 — The bottom line

CORE LESSON

Stop trusting accuracy as a shortcut.

→ Always inspect the confusion matrix

→ Always report precision, recall, or F1

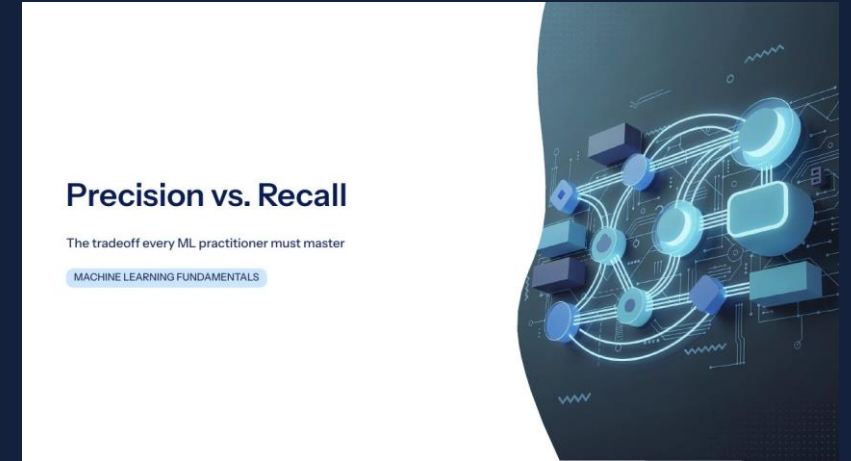
→ Match your metric to what the business can't afford to miss

V19, page 17 — Core lesson

Block 4 of 13, 15 minutes. Video V20.

Precision, recall and the dial between them

The model never says yes or no. It says "73% sure", and you decide where "sure enough" begins.



the deck you watched: V20, Precision vs recall

The one idea

The model never says yes or no. It says "73% sure", and you decide where "sure enough" begins.

Precision

Of everything we flagged, how much was right? Trust in the flags.

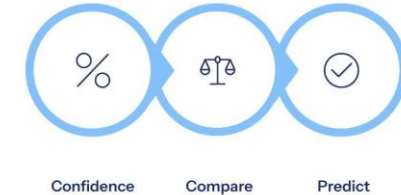
Recall

Of all the real cases, how many did we catch? Thoroughness.

Raise the line and the flags get more trustworthy but you miss more. Lower it and you catch more but annoy more. The line is called the threshold, and nobody chose 0.5 for your business.

FROM VIDEO V20

The Secret: The Threshold



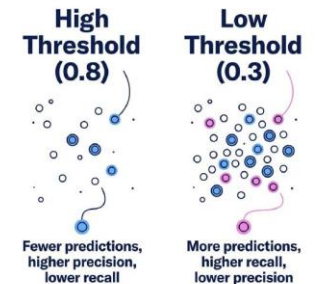
Your model doesn't say "yes" or "no" — it says "I'm 73% confident." The threshold decides what counts as yes.

V20, page 7 — The secret: the threshold

Moving the Dial

The default 0.5 threshold is **arbitrary**. It ignores your business entirely.

Raise it → model becomes **pickier**
Lower it → model casts a **wider net**



V20, page 8 — Moving the dial

A story: two bouncers at the same club



- 1** A club hires a bouncer to keep troublemakers out. "Trouble" is rare: about one guest in twenty.
- 2** Bouncer A turns away anyone who looks even slightly off. Almost no trouble gets in. Also, a lot of good customers spend Friday elsewhere.
- 3** Bouncer B only stops the obvious cases. Everyone he stops was real trouble. Some trouble walks straight in.
- 4** The owner has to choose. A fight costs a licence; an empty club costs the rent. Which mistake is cheaper this month?
- 5** Lesson: A has high recall and low precision, B the reverse. Neither is "better". The owner sets the dial by the price of each mistake.

If you remember nothing else

Three things to keep

- 1 Precision: of our flags, how many were right. Recall: of the real cases, how many we caught.
- 2 The threshold is a dial you can move after the model is built. Up: pickier. Down: wider net.
- 3 Where the dial sits is a business decision. Which mistake costs more?

FROM VIDEO V20

Key Takeaways

- 1 **Precision = Trustworthiness**
Are your positive predictions correct?
- 2 **Recall = Thoroughness**
Did you catch all real positives?
- 3 **Threshold is a Dial**
Move it after your model is built — no retraining needed
- 4 **Never Use Defaults**
0.5 ignores your business. Use business logic instead.

V20, page 16 — Key takeaways

Question 1 of 3

The fraud system flags 80 payments. The team checks them: 60 were real fraud. That week 100 frauds actually happened. What are precision and recall?

- A** Precision 60%, recall 75%
- B** Precision 75%, recall 60%
- C** Both 60%
- D** Both 75%

Hint. Precision: divide by the flags. Recall: divide by the real cases.

Answer your letter in the Zoom chat. We work through the answer together in the session.

FROM VIDEO V20

Precision

Of all positive predictions made — how many were actually correct?

$$\text{Precision} = \frac{\text{True Positives}}{\text{True Positives} + \text{False Positives}}$$

 80 Alerts Raised

System flags 80 transactions as fraud

 60 Are Real Fraud

Team confirms 60 are genuine

75% Precision

$60 \div 80 = 0.75$

V20, page 4 — Precision

Question 2 of 3

The bank lowers its threshold from 0.5 to 0.3. What happens, before anything is retrained?

- A** More frauds caught, more honest customers blocked
- B** Fewer frauds caught, fewer honest customers blocked
- C** Nothing until the model is retrained
- D** Accuracy rises, everything else stays the same

Hint. Lower line, wider net.

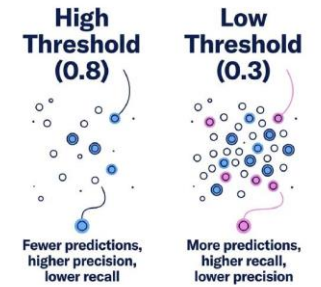
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FROM VIDEO V20

Moving the Dial

The default 0.5 threshold is **arbitrary**. It ignores your business entirely.

Raise it → model becomes **pickier**
Lower it → model casts a **wider net**



V20, page 8 — Moving the dial

Question 3 of 3

Which business should raise its threshold and favour precision?

- A** A hospital screening for a serious disease
- B** A marketing team sending a costly printed catalogue to likely buyers
- C** A bank hunting fraud
- D** A security team flagging possible break-ins

Hint. Where is the false alarm the expensive mistake?

Answer your letter in the Zoom chat. We work through the answer together in the session.

FROM VIDEO V20



Cost Drives the Decision

Every use case has an asymmetry between the cost of a false alarm vs. a missed case.

- **False negative costly?** → Lower threshold, prioritize recall
- **False positive costly?** → Raise threshold, prioritize precision

V20, page 13 — Cost drives the decision

Your turn: move the dial

1,000 payments, 90 of them fraud. Each has a fraud score from the model. Payments above the line are flagged. Move the line; precision and recall are recomputed from the four boxes.

0.50 flag a payment when the score is above this



80% precision **89%** recall

Of 100 flagged, 80 were fraud (precision). Of 90 real frauds, 80 were caught (recall). 10 got through, 20 honest customers were bothered.

	Really fraud	Really honest
Model said fraud	80 caught	20 false alarm
Model said honest	10 missed	890 correctly cleared

Try: 0.20, then 0.80. Which way do precision and recall move? Find the line you would choose for a bank, and the line for a hospital.

Before we move on

Three things from this block

- 1 Precision is trust in the flags; recall is coverage of the real cases.
- 2 The threshold is a dial, and 0.5 is nobody's decision.
- 3 Price the two mistakes, then set the dial.

FROM VIDEO V20



Precision & Recall — Mastered

-  **Precision**
Quality of predictions
-  **Recall**
Coverage of real cases
-  **Threshold**
Your tuning dial
-  **Business Cost**
The ultimate guide

V20, page 20 — Precision and recall, mastered

Don't use defaults.

Use business logic. Your threshold is a strategic asset — treat it like one.

V20, page 19 — Use business logic, not defaults

Block 5 of 13, 12 minutes. Video V21.

F1: one number that cannot hide a weakness

F1 is one score that only looks good when precision and recall are both good.



the deck you watched: V21, F1 score: one number for the whole truth

The one idea

F1 is one score that only looks good when precision and recall are both good.

Plain average

90% precision and 10% recall average to 50%.
Looks like a middling model.

F1

Same numbers give an F1 of 18%. The weak side drags it down, which is the point.

F1 is called a harmonic mean. You never need the formula; you need the behaviour: if either side collapses, F1 collapses.

FROM VIDEO V21

Why Not Just Average?

50%

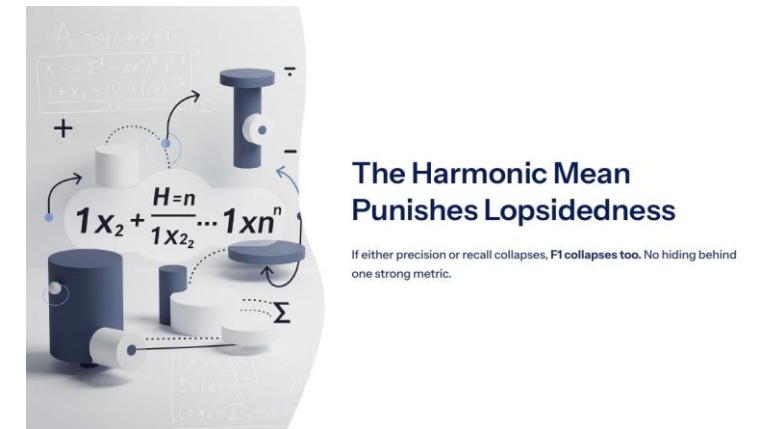
Arithmetic Average
90% precision + 10% recall

18%

F1 Score
Same inputs — harmonic mean

❑ The gap reveals just how **broken a lopsided model** really is. Arithmetic averages hide it.

V21, page 7 — Why not just average?



V21, page 8 — The harmonic mean punishes lopsidedness

A story: the round trip that fooled the average



- 1 You drive 60 km to a client at 60 km/h. One hour.
- 2 Roadworks on the way home. 60 km at 20 km/h. Three hours.
- 3 Average speed for the trip? Most people say 40. The plain average of 60 and 20.
- 4 120 km in four hours is 30 km/h. The slow leg dominates, because you spent three of the four hours in it.
- 5 Lesson: when two rates must both be good, the weak one drags the result down. That is the harmonic mean, and it is exactly how F1 treats precision and recall.

If you remember nothing else

Three things to keep

- 1 F1 punishes lopsided models; the plain average hides them.
- 2 Use F1 when both mistakes hurt about equally. When one mistake is far costlier, look at precision and recall separately.
- 3 F1 is a summary, not a diagnosis: it tells you where to look, then you open the four boxes.

FROM VIDEO V21

Key Takeaways

01

Accuracy lies on imbalanced data

F1 is built for this problem

02

Harmonic mean punishes lopsidedness

Both precision and recall must be high

03

Match your metric to your cost structure

F1, recall, or precision — context decides

04

Always pair F1 with a confusion matrix

A score is a summary, not a verdict



V21, page 15 — Key takeaways

Question 1 of 3

Model A: precision 50%, recall 62.5%, F1 about 0.56. Model B: precision 90%, recall 56%, F1 about 0.68. Model A catches slightly more fraud. Which is the better business model, and why?

- A** A, it catches more fraud
- B** B, its flags are worth acting on
- C** A, lower precision does not matter
- D** Neither, F1 is too low for both

Hint. Think about the analyst who has to check every flag.

Answer your letter in the Zoom chat. We work through the answer together in the session.

FROM VIDEO V21

Two Real Models: Head to Head

Model A	Model B
• Precision: 50% • Recall: 62.5% • F1 $\approx$ 0.56	• Precision: 90% • Recall: 56% • F1 $\approx$ 0.68 ✓

Model B wins — even though it catches *fewer* frauds total. It's tighter, more actionable.

V21, page 9 — Two real models, head to head

Question 2 of 3

A model has precision 100% and recall 2%. Its plain average is 51%.
Roughly what is its F1?

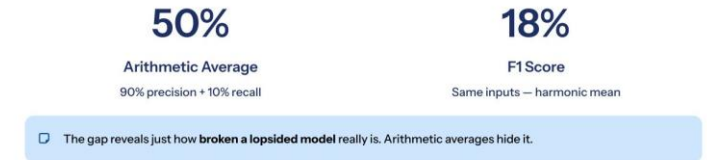
- A** About 51%
- B** About 25%
- C** About 4%
- D** About 98%

Hint. F1 lives near the weaker number.

Answer your letter in the Zoom chat. We work through the answer together in the session.

FROM VIDEO V21

Why Not Just Average?



V21, page 7 — Why not just average?

Question 3 of 3

Cancer screening: a missed case is far worse than a false alarm. The team proposes choosing between models by F1. Good idea?

- A** Yes, F1 balances both mistakes
- B** No, look at recall on its own and accept lower precision
- C** No, use accuracy instead
- D** Yes, as long as F1 is above 0.5

Hint. Are the two mistakes equally bad here?

Answer your letter in the Zoom chat. We work through the answer together in the session.

FROM VIDEO V21



When Recall Matters More

Missing a real cancer case is far worse than a false alarm.

In cancer screening — evaluate **precision** and **recall** separately.
Understand the trade-off before choosing your metric.

V21, page 12 — When recall matters more

Your turn: average versus F1

Set precision and recall by hand. The plain average and F1 are both shown. Make one side collapse and watch which number notices.

90 % precision



10 % recall



50% plain average

18% F1

One side has collapsed. The average still looks respectable; F1 does not.

Try: 100 and 2 (the model that finds almost nothing), 50 and 50, then 90 and 90. F1 sits near the weaker number; the average does not care.

Before we move on

Three things from this block

- 1 The plain average hides a broken model; F1 exposes it.
- 2 Use F1 when both mistakes hurt about equally.
- 3 A score is a summary. The four boxes are the diagnosis.

FROM VIDEO V21

Key Takeaways

01

Accuracy lies on imbalanced data

F1 is built for this problem

02

Harmonic mean punishes lopsidedness

Both precision and recall must be high

03

Match your metric to your cost structure

F1, recall, or precision — context decides

04

Always pair F1 with a confusion matrix

A score is a summary, not a verdict



V21, page 15 — Key takeaways

F1 is a Summary, Not a Diagnosis

⚠ An F1 of 0.7 doesn't tell you *what to do*. It tells you where to look.



Confusion Matrix

Reveals which specific errors your model makes



Precision-Recall Curve

Full context across decision thresholds

Pair F1 with these tools to move from a **number** to a **decision**.

V21, page 14 — F1 is a summary, not a diagnosis

Block 6 of 13, 15 minutes. Video V22.

Turning the threshold into a business decision

Metrics live in percentages. Losses live in money. The threshold is where they meet.



the deck you watched: V22, Translating metric thresholds into business decisions

The one idea

Metrics live in percentages. Losses live in money. The threshold is where they meet.

A miss

50 missed frauds a month at 8,000 each:
400,000 gone.

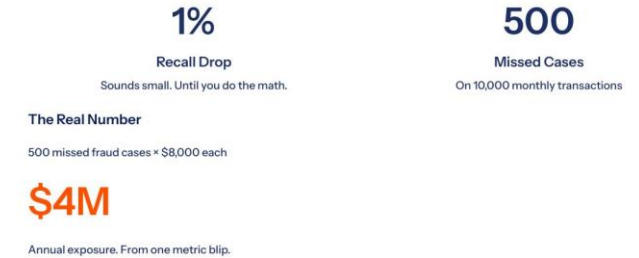
A false alarm

A wrongly declined customer may leave for good, and regulators notice patterns.

Put a price on each mistake, try five thresholds, and write down what each one costs. Then the meeting is about money, which everyone in the room understands.

FROM VIDEO V22

Metrics Live in Percentages. Losses Live in Dollars.

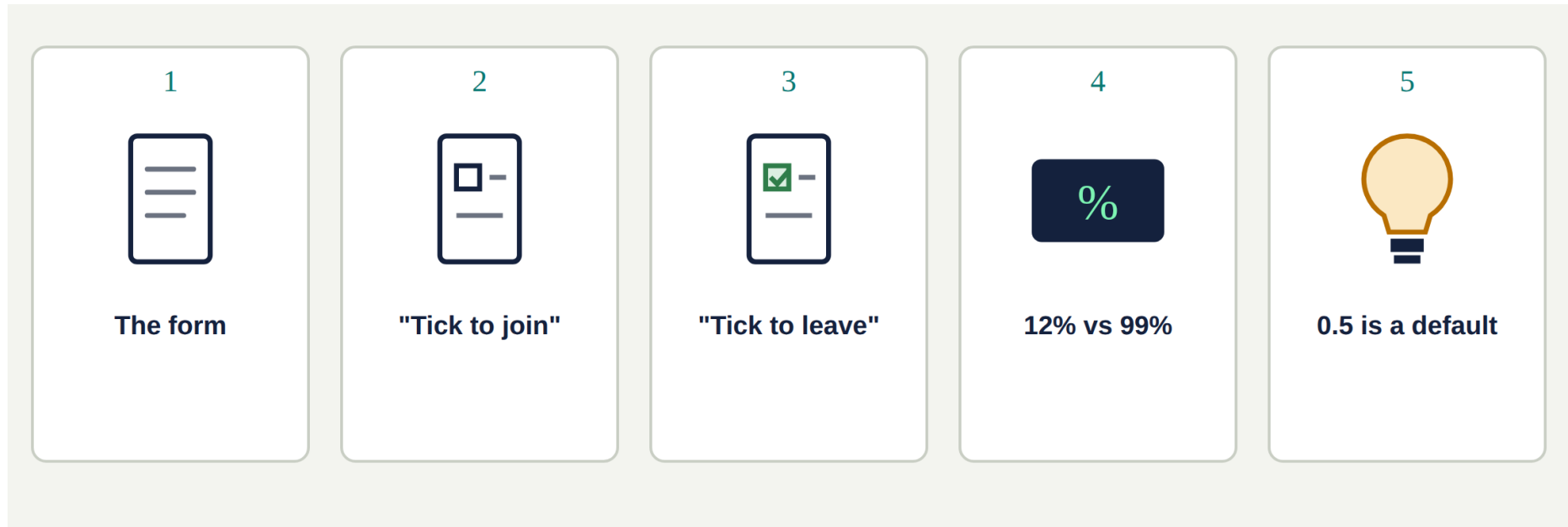


V22, page 6 — Metrics live in percentages, losses live in dollars



V22, page 7 — The cost of a false negative

A story: two countries, one tick box



- 1** Two neighbouring countries ask citizens the same question: will you be an organ donor? Same culture, same values, one form each.
- 2** In the first country the box says: tick here to join. About 12 in 100 tick it.
- 3** In the second country the box says: tick here to leave. About 99 in 100 stay in.
- 4** Same people, same question. The only difference is what happens if you do nothing. The default made the decision for millions.
- 5** Lesson: a default is a decision somebody made for you. Your model's 0.5 threshold is one. Nobody in your company chose it, and it is quietly deciding who gets blocked.

If you remember nothing else

Three things to keep

- 1 0.5 is a choice, not a law. Somebody has to own it.
- 2 Volume times cost per error turns a percentage into a figure the finance team can read.
- 3 The threshold is a policy: five scenarios, a table, a signature, a yearly review.

FROM VIDEO V22

Key Takeaways

- | | |
|---|--|
| 1 0.5 is a choice, not a law
Question every default threshold | 2 Translate metrics to dollars
Volume × cost per error type |
| 3 Run scenario analysis
3-5 thresholds, full business impact table | 4 Govern the decision
Document, sign off, review annually |

V22, page 23 — Key takeaways

Question 1 of 3

The team says recall dropped by 1 point, "a small blip". The bank sees 10,000 payments a month and a missed fraud costs 8,000. How much is that blip worth per year?

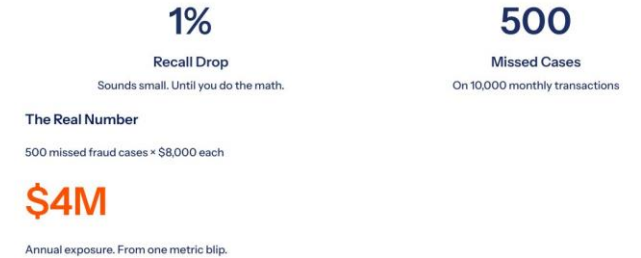
- A About 8,000
- B About 80,000
- C About 800,000
- D About 9.6 million

Hint. One percent of 10,000 payments, times the price, times twelve.

Answer your letter in the Zoom chat. We work through the answer together in the session.

FROM VIDEO V22

Metrics Live in Percentages. Losses Live in Dollars.



V22, page 6 — Metrics live in percentages, losses live in dollars

Question 2 of 3

A credit model's threshold is set high "to be safe". It denies 500 qualified borrowers a year; each would have been worth about 3,000 over their lifetime. What is the threshold costing?

- A** Nothing, safety is free
- B** About 150,000 a year
- C** About 1.5 million a year
- D** It cannot be estimated

Hint. Count times value.

Answer your letter in the Zoom chat. We work through the answer together in the session.

FROM VIDEO V22

Credit Approval: A Real Scenario

The Situation Threshold too high → 500 qualified borrowers denied per year	The Lesson Your threshold isn't protecting you. It's charging you.
The Cost 500 × \$3,000 lifetime value \$1.5M Definite revenue. Walking out the door.	☐ Not someday. Not maybe. Revenue leaving now — because of a number nobody reviewed.

V22, page 9 — Credit approval: a real scenario

Question 3 of 3

The scenario table shows threshold 0.45 is the cheapest overall. The operations manager says her team cannot handle the extra false alarms. What should happen?

- A** Deploy 0.45, the numbers decide
- B** Pick the cheapest threshold the whole business can actually live with, and write down why
- C** Keep 0.5, it is the default
- D** Let the data team decide alone

Hint. Who else has to live with the choice?

Answer your letter in the Zoom chat. We work through the answer together in the session.

FROM VIDEO V22

Don't Just Chase the Cheapest Scenario

→ Can compliance tolerate that fraud miss rate? → Can ops handle the surge in false denials? → Does legal see disparate impact risk?

Cheapest isn't always right. Constraints matter.

V22, page 13 — Do not just chase the cheapest scenario

Your turn: five thresholds, in money

Same 1,000 payments and 90 frauds as before. A missed fraud costs 8,000. Set what a false alarm costs your business, and the table prices five thresholds at once.

500 cost of one false alarm

0.50

cheapest threshold when a false alarm costs 500: total 90,000 per 1,000 payments

THRESHOLD	FRAUDS MISSED	FALSE ALARMS	COST OF MISSES	COST OF ALARMS	TOTAL
0.30	6	198	48,000	99,000	147,000
0.40	9	102	72,000	51,000	123,000
0.50	10	20	80,000	10,000	90,000
0.60	30	14	240,000	7,000	247,000
0.70	48	11	384,000	5,500	389,500

Try: a false alarm at 0 (nobody minds), at 500 (a phone call), at 3,000 (a customer lost for good). Watch the cheapest row move. Then ask: can operations handle the false alarms in that row?

Before we move on

Three things from this block

- 1 Translate every metric into volume times cost per error.
- 2 Try five thresholds, show the money for each, and let the room choose.
- 3 Document, sign, review yearly. The threshold is a living policy.

FROM VIDEO V22

Key Takeaways

- | | |
|---|--|
| 1 0.5 is a choice, not a law
Question every default threshold | 2 Translate metrics to dollars
Volume × cost per error type |
| 3 Run scenario analysis
3-5 thresholds, full business impact table | 4 Govern the decision
Document, sign off, review annually |

V22, page 23 — Key takeaways

Document. Sign Off. Govern.



- Why it matters
- Future you needs to explain the choice
 - Auditors review the decision trail
 - Regulators expect a paper trail

☒ Threshold selection is governance — not a notebook tweak.

V22, page 15 — Document, sign off, govern

Block 7 of 13, 12 minutes. Video V23.

Saying it so the room can decide

A model is worth exactly the decisions it changes. Nobody decides on a number they do not understand.



the deck you watched: V23, Model evaluation as a stakeholder skill

The one idea

A model is worth exactly the decisions it changes.
Nobody decides on a number they do not understand.

What you say

"Recall is 87%, precision 96%, F1 0.91."

What they can use

"We catch 87 frauds in every 100, and flag 4 honest payments for every real fraud we find."

Same numbers. The second version can be argued with, priced, and decided on. The first sits on a shelf.

FROM VIDEO V23



The Communication Failure

You say:

- "Precision is 96%"
- "Recall is 87%"
- "F1 score is 0.91"

They hear:

- "How many frauds slip through?"
- "How many real customers get blocked?"
- "What does this cost us?"

V23, page 4 — The communication failure

Translation in Action: Fraud Detection

✗ Statistical Language

"Recall is 87%"

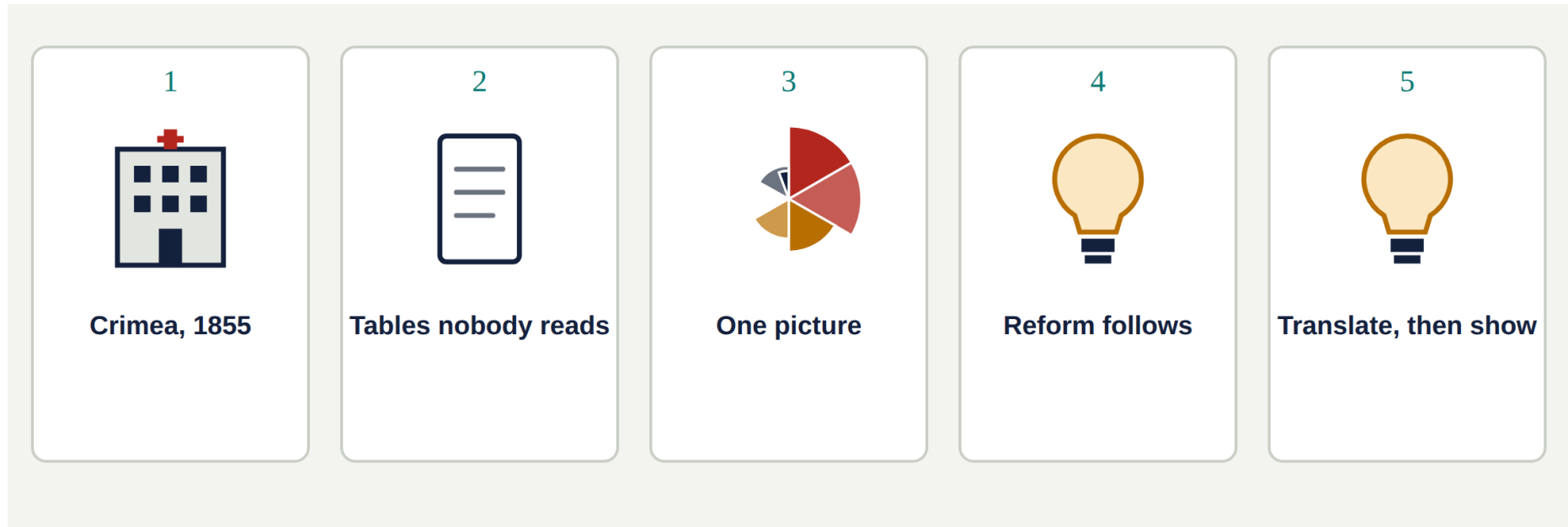
✓ Business Language

"We catch 87 frauds per 100 — and flag 4 legitimate transactions for every real fraud we find."

Same data. Completely different clarity.

V23, page 6 — Translation in action: fraud detection

A story: the nurse who won an argument with a picture



- 1** A military hospital in the Crimean War. Soldiers are dying in their beds far more than on the battlefield. Florence Nightingale is counting them.
- 2** She sends the numbers to London in tables. Columns of deaths by month and cause. The generals do not read tables.
- 3** So she draws a picture: a wheel with a wedge for each month, coloured by cause. The disease wedges dwarf the battle wounds. You can see it from across the room.
- 4** Parliament and the army act. Sanitation is overhauled. The chart did what the tables could not.
- 5** Lesson: the numbers were right all along. They changed nothing until they were translated into a form the decision-makers could see.

If you remember nothing else

Three things to keep

- 1 Every metric must map to a business consequence, or do not say it.
- 2 Lead with the one number your listener decides on: risk teams want misses, customer teams want false alarms, finance wants money.
- 3 Show the trade-off, benefit and cost together, up front.

FROM VIDEO V23

CORE PRINCIPLE

**Every metric must map
to a business
consequence.**

V23, page 5 — Every metric must map to a business consequence

THE GOLDEN QUESTION

**"What decision will this metric
influence — and who makes it?"**

Answer this before every stakeholder meeting.

V23, page 10 — The golden question

Question 1 of 3

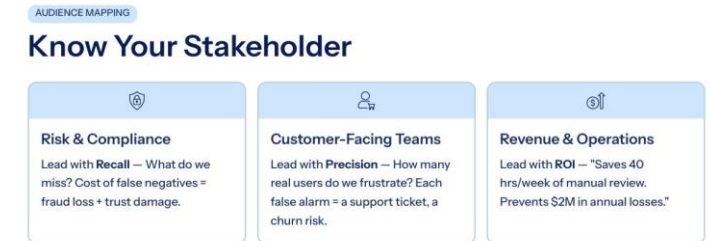
You are presenting the fraud model to the head of customer service.
Which opening line lands?

- A** "Precision is 96% and recall is 87%."
- B** "Accuracy is 99.2%."
- C** "Four honest customers get blocked for every real fraud we stop. Here is what that costs your team per week."
- D** "The F1 score is 0.91, up from 0.85."

Hint. What decision does she make?

Answer your letter in the Zoom chat. We work through the answer together in the session.

FROM VIDEO V23



V23, page 8 — Know your stakeholder

Question 2 of 3

The model has a real downside: it will block about 1 in 25 honest payments. When do you mention it?

- A** Only if someone asks
- B** After launch, once the benefits are visible
- C** In the same slide as the benefit, before the decision
- D** Never; it would kill the project

Hint. What erodes trust: a known cost, or a discovered one?

Answer your letter in the Zoom chat. We work through the answer together in the session.

FROM VIDEO V23

Don't Hide the Downside



Stakeholders expect trade-offs.

What erodes trust is discovering the costs **after** the fact.

Show the full picture — benefit *and* cost — together, upfront.

☒ Transparency builds credibility.
Omission destroys it.

V23, page 17 — Do not hide the downside

Question 3 of 3

Same model, three meetings. Risk and compliance, then the customer team, then finance. What changes between the three?

- A** The model
- B** The numbers
- C** Which number you lead with, and what you translate it into
- D** Nothing; consistency matters

Hint. Who is in the room, and what do they decide?

Answer your letter in the Zoom chat. We work through the answer together in the session.

FROM VIDEO V23



One Model, Three Translations

The same underlying metrics tell completely different stories depending on who's listening.

The skill isn't building the model. It's knowing which story each stakeholder needs to hear.

V23, page 9 — One model, three translations

Your turn: same model, different room

The four boxes from the fraud model: 80 caught, 20 honest blocked, 10 missed, 890 cleared, per 1,000 payments. Pick who is in the room and read the sentence they can decide on.

Who is in the room?

Risk and compliance



Same numbers every time. Only the first sentence changes.

What risk and compliance hears

We miss 10 frauds in every 1,000 payments: one fraud in nine gets through.

At 8,000 per missed fraud that is 80,000 of loss per 1,000 payments.

the decision this feeds

Is a one-in-nine miss rate acceptable to compliance?

Try: read the data-team version out loud, then the finance version. Which one could you say yes or no to?

Before we move on

Three things from this block

- 1 Metric, plain words, consequence: keep that glossary next to you.
- 2 Lead with the number your listener decides on.
- 3 Benefit and cost on the same slide, before the decision.

FROM VIDEO V23

FINAL TAKEAWAY

**Model evaluation is a
communication skill first.
A technical skill second.**

Every metric tells a business story. Your job is to translate it — not just calculate it.

V23, page 25 — Final takeaway

What Great Translators Do



Know Their Audience

Stakeholder role and incentives determine which metric leads.



Frame Both Sides

Benefit and cost together — never just the upside.



Make It Visible

Strip jargon. Annotate with outcomes. One story, one chart.



Invite the Decision

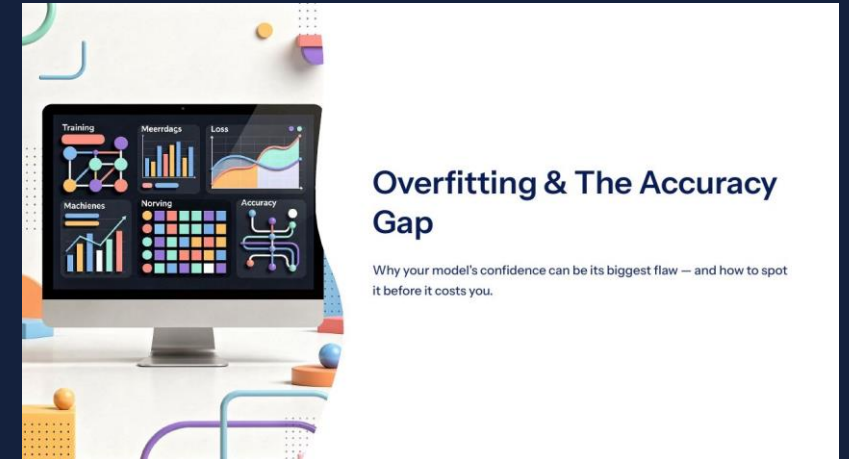
Present options. Respect authority. Let stakeholders lead.

V23, page 22 — What great translators do

Block 8 of 13, 15 minutes. Video V24.

Overfitting and the gap that gives it away

A model that memorises its homework aces the homework and fails the exam.



the deck you watched: V24, Overfitting and the accuracy gap

The one idea

A model that memorises its homework aces the homework and fails the exam.

Training score

How well the model does on the data it learned from. 95%.

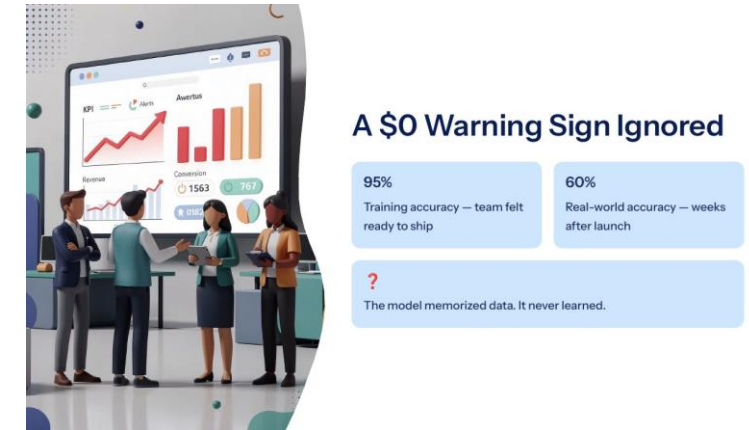
Validation score

How well it does on data it has never seen.
60%.

The 35-point gap is the diagnosis. The model learned the quirks of one dataset, not the pattern underneath. The gap was visible before launch; nobody looked.

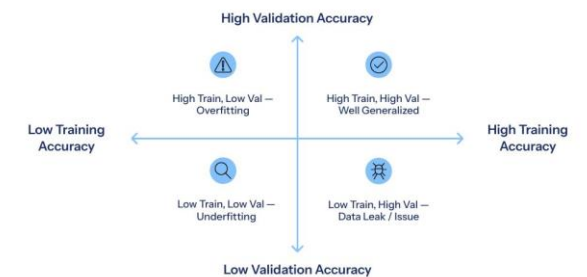
Block 8 of 13: Overfitting

FROM VIDEO V24



V24, page 2 — A warning sign ignored

Training vs. Validation Accuracy

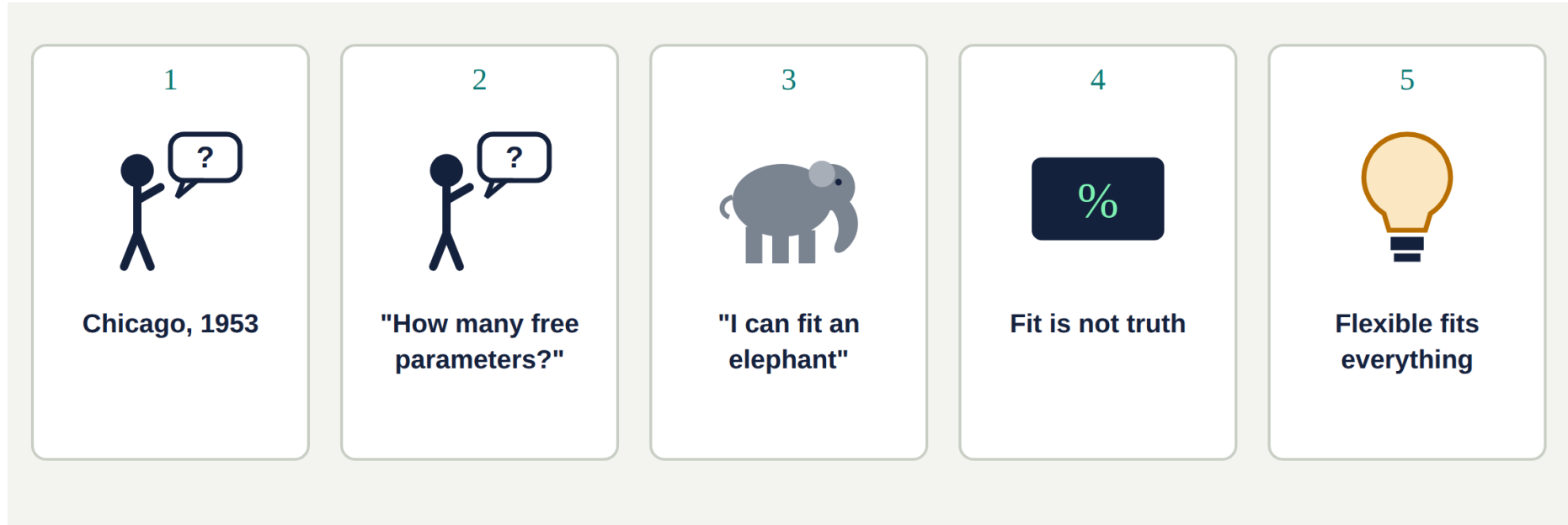


The gap between these two numbers is your most important diagnostic signal. A wide gap means overfitting.

V24, page 4 — Training vs validation accuracy

Machine Learning and AI in Business · Session 2 · 10 September 2026

A story: the elephant with four parameters



- 1** A young physicist, Freeman Dyson, takes a new theory to Enrico Fermi. The theory's curve matches the experimental data beautifully.
- 2** Fermi barely looks at the curve. He asks how many adjustable numbers Dyson tuned to make it fit. Four, says Dyson.
- 3** Fermi quotes his friend von Neumann: with four parameters I can fit an elephant, and with five I can make him wiggle his trunk.
- 4** Dyson's beautiful match proved nothing. Enough adjustable parts can be bent to fit any data, including noise. The theory was abandoned.
- 5** Lesson: a flexible enough model fits anything you show it. The only honest test is data it has never seen.

If you remember nothing else

Three things to keep

- 1 Training score tells you how well the model fits what it saw. Validation score tells you whether it learned anything.
- 2 A gap of more than 5 to 10 points is a red flag. Do not rationalise it; investigate.
- 3 Complex model plus small data is the recipe for memorising noise.

FROM VIDEO V24

Don't trust training accuracy.

Trust the **gap**. It tells the real story.

V24, page 14 — Trust the gap

Spot the Gap



V24, page 12 — Spot the gap

Question 1 of 3

A model scores 97% on training data and 71% on validation data. What is the most likely diagnosis?

- A** Underfitting: the model is too simple
- B** Overfitting: the model memorised the training data
- C** A healthy model with a tough validation set
- D** The validation set is too small to matter

Hint. Look at the gap, not the level.

Answer your letter in the Zoom chat. We work through the answer together in the session.

FROM VIDEO V24

Training vs. Validation Accuracy



The gap between these two numbers is your most important diagnostic signal. A wide gap means overfitting.

V24, page 4 — Training vs validation accuracy

Question 2 of 3

Which of these is a pattern worth keeping, rather than noise?

- A** "Customers whose surname starts with a vowel spend more" (true in this year's data)
- B** "Customers with a longer engagement history spend more" (true in this year's data and last year's)
- C** "Customers who signed up on a Tuesday cancel less" (true in the training sample only)
- D** "The 300th customer in the file was a big spender"

Hint. Which one would you bet holds next year?

Answer your letter in the Zoom chat. We work through the answer together in the session.

FROM VIDEO V24

Noise vs. Signal

✗ Noise (Overfit) "Customers with last names starting with vowels spend more." True in training. Gone in the real world.	✓ Signal (Generalized) "Customers with longer engagement history predict higher spending." Holds in training. Holds on new customers. That's a real pattern.
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V24, page 8 — Noise vs signal

Question 3 of 3

The data science team wants to ship a model with a 92% training score and a 68% validation score because "the validation set is probably unlucky". What should the business ask for first?

- A** Ship it and monitor
- B** A different, larger validation set and the learning curves, before any deployment
- C** More features to lift the validation score
- D** A more complex model

Hint. What does the gap cost to check now, versus after launch?

FROM VIDEO V24

Spot the Gap



The Rule of Thumb

A gap of **more than 5–10 percentage points** is a red flag.
Don't rationalize it. Don't deploy. **Investigate.**

V24, page 12 — Spot the gap

Answer your letter in the Zoom chat. We work through the answer together in the session.

Your turn: read the gap

Slide the model's complexity (tree depth 1 to 10). Training error always falls. Validation error is the one that predicts the real world. Watch the gap.

4 tree depth (complexity)

15% training error

15% validation error

Gap: 0 points. Zone 2, the sweet spot:
validation error at its lowest, small
gap. Ship this.



Try: depth 1, depth 4, depth 7. At which depth does the gap pass 10 points? What would you say to a team proposing depth 9?

Before we move on

Three things from this block

- 1 Trust the gap, not the training score.
- 2 Signal survives new data; noise does not.
- 3 Checking the gap costs hours. Missing it costs the launch.

FROM VIDEO V24

Don't trust training accuracy.

Trust the **gap**. It tells the real story.

V24, page 14 — Trust the gap

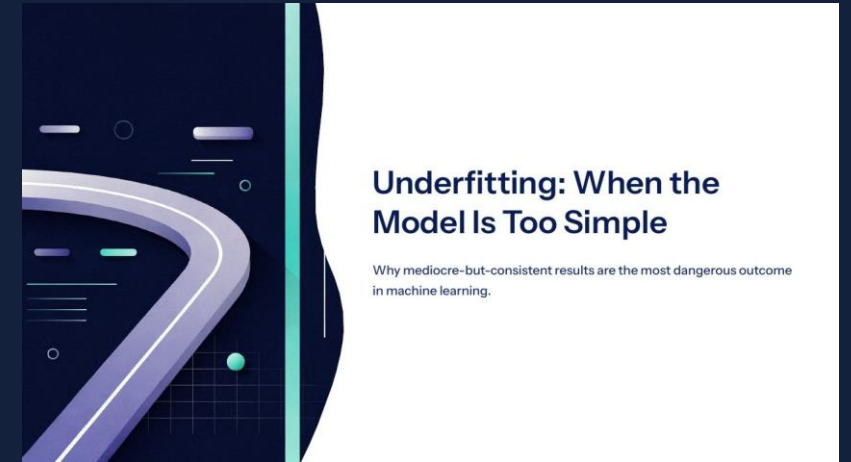


V24, page 15 — What is next: from diagnosis to cure

Block 9 of 13, 12 minutes. Video V25.

Underfitting: the model that is too simple to see

The dangerous failure is the quiet one: mediocre everywhere, alarming nowhere.



the deck you watched: V25, Underfitting: when the model is too simple

The one idea

The dangerous failure is the quiet one: mediocre everywhere, alarming nowhere.

Training 78%

The model cannot even fit the data it learned from.

Validation 78%

Same on new data. No gap. Nothing looks broken.

Equal and mediocre on both sides is the underfitting signature. It clears the deployment bar and fails slowly, silently and expensively.

FROM VIDEO V25

The Underfitting Signature

Training Accuracy	Validation Accuracy
78%	78%

Equal and mediocre on **both sides** — that's the underfitting signature. No gap. Just persistent weakness.

V25, page 5 — The underfitting signature

Why Teams Miss It

- Validation accuracy looks "reasonable" at 78%
- Nobody checks that training accuracy is *also* 78%
- Consistent mediocrity reads as acceptable — and gets deployed



V25, page 6 — Why teams miss it

A story: the innkeeper with one bed



- 1 In a Greek legend an innkeeper named Procrustes offers every traveller a bed. He has exactly one size.
- 2 A tall guest overhangs the bed, so Procrustes cuts him down to size.
- 3 A short guest is stretched until he fits. Every guest, by morning, fits the bed perfectly.
- 4 The inn's records are spotless: every guest fitted. The guests are another matter.
- 5 Lesson: a model too simple does not fit the data; it forces the data to fit it. A straight line through a curved trend cuts off the peaks and stretches the valleys.

If you remember nothing else

Three things to keep

- 1 Underfitting is low scores on training and validation alike. The warning sign is the level, not the gap.
- 2 Causes: the wrong kind of model, too few inputs, training stopped early, constraints too tight.
- 3 The fastest fix is usually a missing input, not more data.

FROM VIDEO V25

The Core Rule

Underfitting = model too **simple**, not too complex.

The warning sign is simultaneous poor performance on training *and* validation — not the gap between them.

V25, page 17 — The core rule

Root Causes of Underfitting



Wrong Model Type

Linear model on a nonlinear problem — straight line on a curved trend



Too Few Features

Critical variables never fed in — the model is flying blind



Early Stopping

Training halted before meaningful patterns are learned



Over-Constrained Architecture

Excessive regularization or too few layers — model can't adapt

V25, page 7 — Root causes of underfitting

Question 1 of 3

A recruitment model uses only job title and years of experience. It scores 61% on training and 62% on validation. What is wrong?

- A** It is overfitting
- B** It is underfitting: it lacks the information to do the job
- C** The validation set is bad
- D** Nothing; 62% is acceptable

Hint. Low on both sides: what is missing?

Answer your letter in the Zoom chat. We work through the answer together in the session.

FROM VIDEO V25

Real Example: Recruitment Classifier



Inputs Used

- Job title
- Years of experience

Ignored

- Industry, location, certifications

Result: 61% training / 62% validation — the model simply lacks the information to do the job.

V25, page 8 — Real example: recruitment classifier

Question 2 of 3

Why does underfitting often reach production when overfitting does not?

- A** Underfitting is rarer
- B** Overfitting shows a big gap that reviewers catch; underfitting shows consistent, unremarkable scores that clear the bar
- C** Underfit models are cheaper to run
- D** Reviewers only check validation scores, and underfit models have none

Hint. Which failure looks alarming in a review meeting?

Answer your letter in the Zoom chat. We work through the answer together in the session.

FROM VIDEO V25



V25, page 10 — Why underfitting is more dangerous than overfitting

Question 3 of 3

A sales forecast is within 15% on training and validation data but misses every seasonal peak. Which fix is most likely to help first?

- A** Collect three more years of the same data
- B** Add inputs that carry the season: month, holidays, promotions
- C** Use a simpler model
- D** Lower the threshold

Hint. What can the model not see?

Answer your letter in the Zoom chat. We work through the answer together in the session.

FROM VIDEO V25

The Hidden Problem

A sales model predicts within 15% error on training and test data. Leadership is satisfied.

But it misses seasonal spikes every quarter.

☐ Underfitting hides as mediocre results everywhere — no alarm bells flashing.



V25, page 2 — The hidden problem

Your turn: give the model something to see

The recruitment model from the video. Tick the inputs it is allowed to use. Training and validation scores update. Notice there is never a gap; there is only a level.

- ☒ Job title
- ☒ Years of experience
- ☐ Industry
- ☐ Location
- ☐ Certifications

61% training accuracy

62% validation accuracy

Low on both sides, no gap: underfitting. The model lacks the information to do the job.

Try: untick everything, then add one input at a time. Which single input moves the score most? That is the one the model was blind to.

Before we move on

Three things from this block

- 1 Underfitting hides as consistent mediocrity.
- 2 Check the level of both scores, not just the gap.
- 3 A missing input is the usual cause and the fastest fix.

FROM VIDEO V25

Key Takeaways

01	02
Underfitting hides as mediocre-but-consistent results No alarm bells — it clears deployment thresholds undetected	Check both training and validation error — the level, not just the gap Both high = underfitting signal
03	04
Fix with complexity: better features, deeper models, longer training The model needs more, not less	In business, quiet underperformance compounds into serious damage Missed risk signals, lost revenue — quarter after quarter

V25, page 20 — Key takeaways

The Core Rule

Underfitting = model too **simple**, not too complex.

The warning sign is simultaneous poor performance on training *and* validation — not the gap between them.

V25, page 17 — The core rule

Block 10 of 13, 12 minutes. Video V26.

Why you never grade your own exam

Testing a model on the data it learned from is grading your own homework.



the deck you watched: V26, Train-test split: why it is non-negotiable

The one idea

Testing a model on the data it learned from is grading your own homework.

Same data for both

Train on everything, test on everything. 95% in the lab, 70% in production.

Hold some back

Lock 20 to 30% away before training. Test on it once. The score you get is the score you will see in the real world.

The test set is the stranger the model has never met. Everything else is a rehearsal.

FROM VIDEO V26



The Self-Grading Problem

The Student Analogy
Grading your own exam = guaranteed perfect score

The ML Equivalent
Training and testing on the same data = fake accuracy

The Real Number
95% in the lab → 70% in production

V26, page 2 — The self-grading problem

The Fix: Train-Test Split



The test set acts as a proxy for real-world data — the model never sees it during training.

V26, page 6 — The fix: train-test split

A story: Clever Hans, the horse who could count



- 1 A horse called Hans answers arithmetic questions by tapping his hoof. Crowds gather. Newspapers are amazed. A commission of experts says: no trick.
- 2 Ask Hans what three plus four is and he taps seven. He passes every test the experts set. Every test where the questioner knows the answer.
- 3 A young psychologist, Oskar Pfungst, tries something new: the person asking does not know the answer. Hans fails almost every time.
- 4 Hans had learned to read the questioner: a tiny relaxing of the head when the right number came. His "arithmetic" was a leak from the examiner.
- 5 Lesson: a test that contains the answer is not a test. The held-out set is the questioner who does not know; if the model still scores, it learned something real.

If you remember nothing else

Three things to keep

- 1 Split before you train, and never touch the test set until the end.
- 2 Shuffle first, so both parts look like the whole.
- 3 Use the test set once. Tune on a separate validation set, or you have leaked the answer.

FROM VIDEO V26

Key Takeaways

- Split before you train
No exceptions — ever
- Shuffle your data
Remove ordering bias before splitting
- Evaluate the test set once
Use validation for iteration, test for truth
- Use cross-validation for small datasets
More signal from the same data
- Document everything
Reproducibility = credibility

V26, page 14 — Key takeaways



Lock It Away — Before You Begin

Set aside 20–30%
Before any training starts

Train on 70–80%
Model learns only from this portion

Evaluate on the rest
One honest look at true performance

V26, page 7 — Lock it away before you begin

Question 1 of 3

A team trains a model on all 10,000 rows, then measures accuracy on the same 10,000 rows: 95%. What does that number tell you about performance on next month's customers?

- A** About 95%
- B** Almost nothing; the model has seen the answers
- C** A bit less, maybe 90%
- D** More than 95%, models improve with use

Hint. Who marked the exam?

Answer your letter in the Zoom chat. We work through the answer together in the session.

FROM VIDEO V26



The Self-Grading Problem

The Student Analogy
Grading your own exam =
guaranteed perfect score

The ML Equivalent
Training and testing on the
same data = fake accuracy

The Real Number
95% in the lab → 70% in production

V26, page 2 — The self-grading problem

Question 2 of 3

The team holds out 20% as a test set, then tunes the model dozens of times, checking the test score each time and keeping the best. What has gone wrong?

- A** Nothing; that is what a test set is for
- B** The test set has been used to choose the model, so its score is now inflated
- C** The test set is too small
- D** The training set is now too small

Hint. How many times can you look at the answer before it stops being a test?

Answer your letter in the Zoom chat. We work through the answer together in the session.

FROM VIDEO V26



The Golden Rule: Use the Test Set Once

- ☒ **Tune hyperparameters on test data**
You've leaked information — accuracy is now inflated
- ☒ **Tune on a validation set**
Keep the test set pristine for your final, honest evaluation

V26, page 12 — The golden rule: use the test set once

Question 3 of 3

The rows in the customer file are sorted by sign-up date. The team takes the first 80% for training and the last 20% for testing, without thinking about it. What is the risk?

- A** None; any split works
- B** The two parts differ systematically (old customers versus new), so the test score is misleading
- C** The test set is too big
- D** The model will overfit the dates

Hint. Do the two parts look like the same population?

Answer your letter in the Zoom chat. We work through the answer together in the session.

FROM VIDEO V26

Always Shuffle First

Why Randomize?

- Removes ordering bias
- Both sets become representative
- Prevents the model learning data-order patterns

Time-based or grouped data? An unshuffled split embeds **systematic bias** you won't even notice.

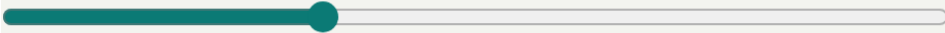


V26, page 9 — Always shuffle first

Your turn: how much to hold back

1,000 customer rows. Choose how much to lock away as the test set. The margin shows how far a test score can be off simply because the test set is small.

20 % held out as the test set



☐ grade on the training rows instead (the self-graded exam)

71% score ± 5.7 points

Measured on 200 rows the model never saw; the model learned from 800. A reasonable balance for this dataset.

Try: 5% held out (a tiny test set: wide margin), then 50% (an honest score, but the model learned from only half the rows). Then tick the self-graded box and see the number the team wanted to report.

Before we move on

Three things from this block

- 1 The test set is the stranger the model has never met.
- 2 Shuffle, split, lock it away, and open it once.
- 3 Cross-validation: five test sets from one dataset when data is scarce.

FROM VIDEO V26

Key Takeaways

- **Split before you train**
No exceptions — ever
- **Shuffle your data**
Remove ordering bias before splitting
- **Evaluate the test set once**
Use validation for iteration, test for truth
- **Use cross-validation for small datasets**
More signal from the same data
- **Document everything**
Reproducibility = credibility

V26, page 14 — Key takeaways

Block 11 of 13, 12 minutes. Video V27.

The validation curve: finding the sweet spot

Two lines on one chart tell you how complex the model should be.



the deck you watched: V27, The validation curve

The one idea

Two lines on one chart tell you how complex the model should be.

Training error

Falls and falls as the model gets more complex. It always looks better.

Validation error

Falls, bottoms out, then rises. The bottom is where you stop.

Three zones: too simple on the left, the sweet spot in the middle, too complex on the right. The gap between the lines is your diagnostic.

FROM VIDEO V27

What Is a Validation Curve?



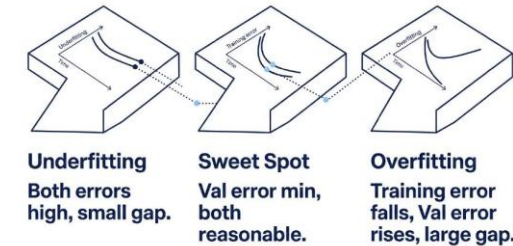
Two lines. One story.

- **Training error** — how well the model fits what it learned
- **Validation error** — how well it generalizes to new data

The gap between them tells you everything.

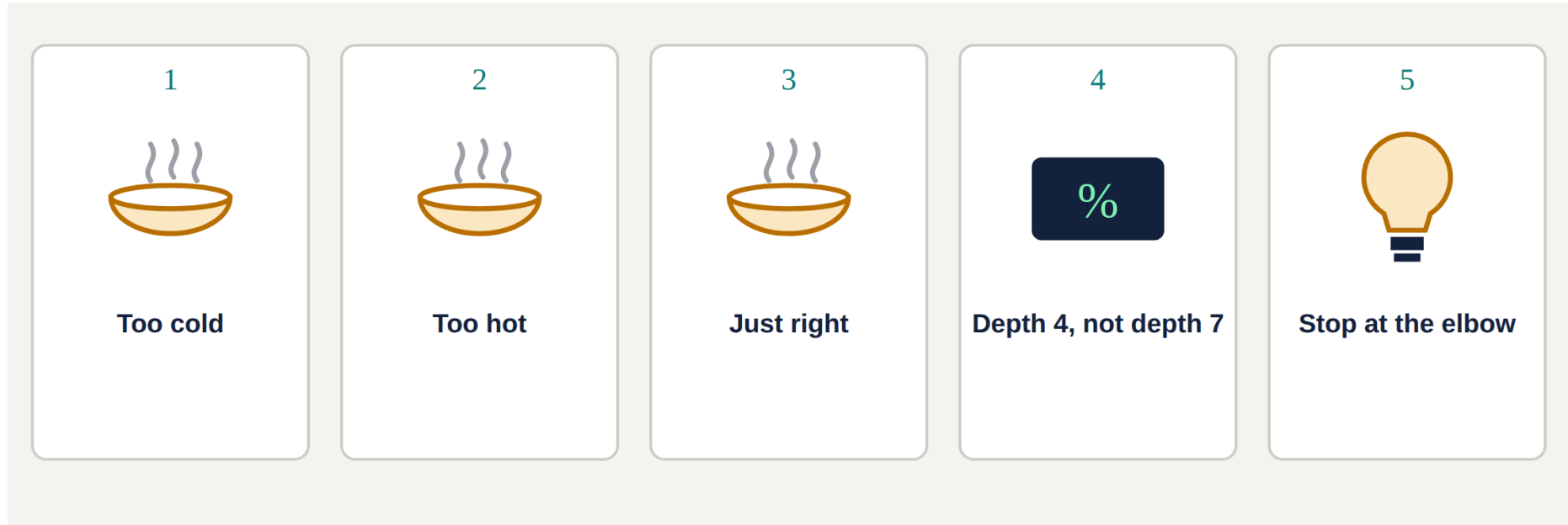
V27, page 5 — What is a validation curve?

Three Zones on Every Validation Curve



V27, page 8 — Three zones on every validation curve

A story you know: three bowls of porridge



- 1 Goldilocks tries the first bowl. Too cold. The model that is too simple: it misses the pattern, and it is wrong on old data and new alike.
- 2 The second bowl burns her mouth. The model that is too complex: perfect on what it saw, useless on anything new.
- 3 The third bowl. Not the coldest, not the hottest: the one that is right for the job.
- 4 The video's tree: depth 1 fails at both, depth 7 memorises, depth 4 is just right at 15% error on new data. The chart finds the bowl for you.
- 5 Lesson: sweep the complexity from low to high, watch the validation error, stop where it stops improving. Then write down why.

If you remember nothing else

Three things to keep

- 1 Training error always falls with complexity; validation error bottoms out. The bottom is the answer.
- 2 Small gap: keep going. Widening gap: stop.
- 3 Pick the setting at or just before the elbow, and document the choice.

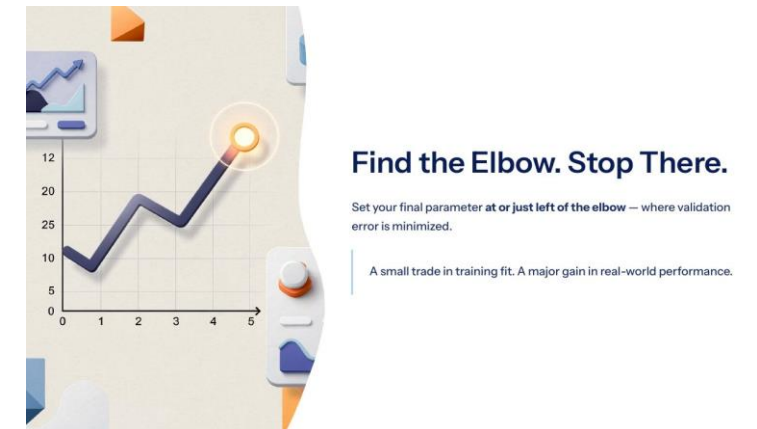
FROM VIDEO V27

The Core Takeaway

Don't chase low training error.

Chase the complexity level where your model **generalizes best** to the world it will actually face.

V27, page 18 — The core takeaway



V27, page 16 — Find the elbow, stop there

Question 1 of 3

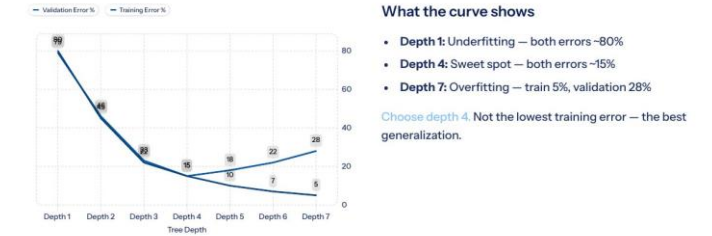
A decision tree: depth 1 gives 80% error on training and validation; depth 4 gives 15% and 15%; depth 7 gives 5% and 28%. Which depth ships?

- A** Depth 7, lowest training error
- B** Depth 4, lowest validation error
- C** Depth 1, simplest is safest
- D** Depth 10, even lower training error

Hint. Which line predicts new data?

FROM VIDEO V27

Real Example: Decision Tree Classifier



V27, page 13 — Real example: decision tree classifier

Answer your letter in the Zoom chat. We work through the answer together in the session.

Question 2 of 3

On the validation curve, training error is still falling and validation error has started rising. What does that tell you?

- A** Add more complexity, the training error is improving
- B** Stop: you have passed the sweet spot and are memorising noise
- C** The data has an error
- D** Both lines will meet if you keep going

Hint. Read the gap.

FROM VIDEO V27

Read the Gap as a Diagnostic

Small Gap

Added complexity helps both training and generalization. Keep going.

Widening Gap

Training error falls — but validation error rises. **Red flag.** Stop here.

V27, page 12 — Read the gap as a diagnostic

Answer your letter in the Zoom chat. We work through the answer together in the session.

Question 3 of 3

The validation curve says the model is overfitting at every setting the team tried. Which second chart tells them whether collecting more data would help?

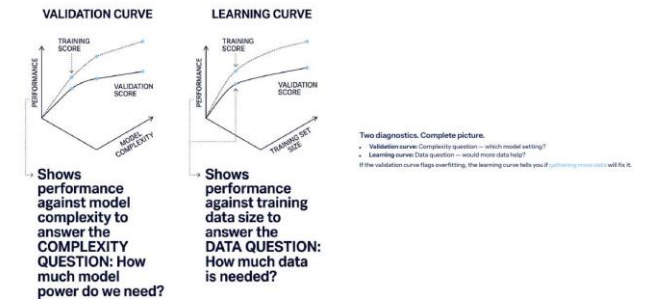
- A** A pie chart of the classes
- B** The learning curve: error against training-set size
- C** A bar chart of feature counts
- D** The confusion matrix

Hint. Complexity question versus data question.

Answer your letter in the Zoom chat. We work through the answer together in the session.

FROM VIDEO V27

Validation Curves vs. Learning Curves

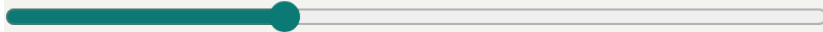


V27, page 19 — Validation curves vs learning curves

Your turn: find the elbow

Slide the tree depth from 1 to 10. Where does validation error stop improving? That is the elbow, and the depth you ship.

4 tree depth (complexity)



15% training error

15% validation error

Zone 2, the sweet spot: validation error at its lowest, small gap. Ship this.



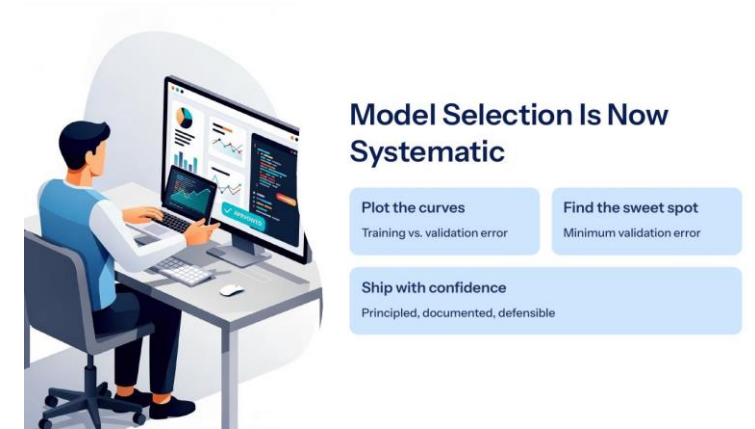
Try: sweep slowly from 1 to 10. Say the zone out loud at each step. Stop at the elbow and write down why.

Before we move on

Three things from this block

- 1 The validation line bottoms out. That bottom is the model.
- 2 Widening gap means stop.
- 3 Two diagnostics: validation curve for the setting, learning curve for more data.

FROM VIDEO V27



V27, page 20 — Model selection is now systematic

The Core Takeaway

Don't chase low training error.

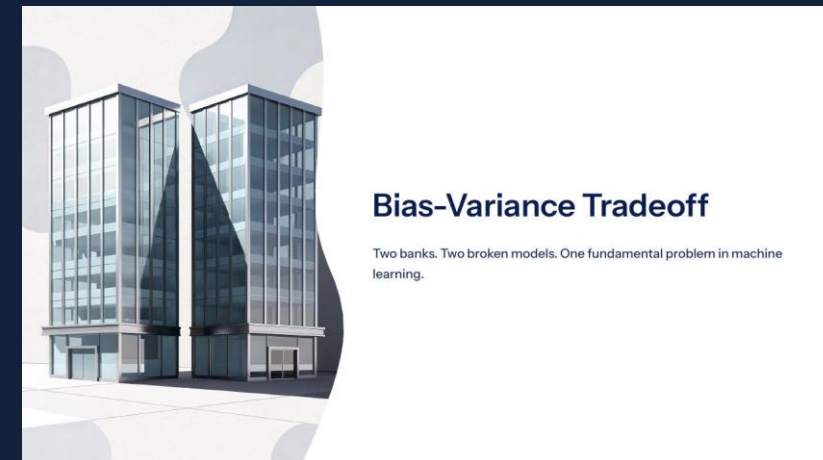
Chase the complexity level where your model **generalizes best** to the world it will actually face.

V27, page 18 — The core takeaway

Block 12 of 13, 12 minutes. Video V28.

Two ways to be wrong: bias and variance

Every model is wrong in one of two ways: too rigid to learn, or too flexible to be trusted.



the deck you watched: V28, Bias-variance trade-off

The one idea

Every model is wrong in one of two ways: too rigid to learn, or too flexible to be trusted.

Bias

Too simple. Consistent, and consistently wrong. The weather model that says 22 degrees every day.

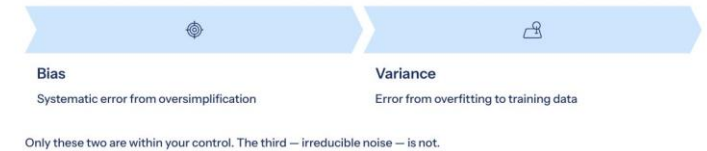
Variance

Too flexible. Perfect on what it saw, swinging wildly on anything new.

You cannot minimise both by accident. Every complexity choice moves one up and the other down. More data is the one thing that helps both.

FROM VIDEO V28

Two Sources of Error



V28, page 3 — Two sources of error

What Is Bias?

A weather model that always predicts 72°F — summer, winter, monsoon.

Consistent. Consistently wrong. The model is **too rigid to learn real patterns**.



V28, page 4 — What is bias?

A story: two archers and a coach



1 Every arrow lands in a tight group, ten centimetres left of the centre. Consistent. Consistently off.

2 Arrows all around the centre, none in it. Some high, some low, some wide. On average, dead centre; on any single shot, anywhere.

3 A has a sight set wrong: a systematic error. B has an unsteady hand: a random one. Same target, two different problems, two different fixes.

4 A: adjust the sight (change the model). B: more practice, more arrows (more data), or a steadier stance (constrain the model).

5 Lesson: A is bias, B is variance. Knowing which one you have tells you what to do next, and the fixes are not interchangeable.

If you remember nothing else

Three things to keep

- 1 Bias: too simple, wrong in the same way every time. Variance: too flexible, wrong in a different way every time.
- 2 Bias shows as low scores everywhere; variance shows as a gap.
- 3 More diverse data cuts variance without adding bias. Collect it first if you can.

FROM VIDEO V28

Key Takeaways

 Bias Too simple → underfitting	 Variance Too complex → overfitting
 More Data Breaks the tradeoff	 Business Risk Drives your final choice

V28, page 17 — Key takeaways

More Data Breaks the Tradeoff.

Adding diverse training examples reduces variance — **without increasing bias.**



V28, page 9 — More data breaks the trade-off

Question 1 of 3

Bank A's loan model rejects 90% of applicants and scores poorly on training data too. Bias or variance, and what is the fix?

- A** Variance; add regularisation
- B** Bias; give the model more inputs and more flexibility
- C** Variance; collect more data
- D** Neither; lower the threshold

Hint. Poor on training data: which archer is that?

Answer your letter in the Zoom chat. We work through the answer together in the session.

FROM VIDEO V28

Bank A: Diagnosing the Problem



V28, page 14 — Bank A: diagnosing the problem

Question 2 of 3

Bank B's fraud model is perfect on training data and flags honest customers daily in production. Bias or variance, and what is the fix?

- A** Bias; add features
- B** Variance; add diverse fraud examples, or constrain the model
- C** Bias; train longer
- D** Variance; use a more complex model

Hint. Perfect on training, erratic in production: which archer?

Answer your letter in the Zoom chat. We work through the answer together in the session.

FROM VIDEO V28

Bank B: Diagnosing the Problem



Flags legitimate transactions daily

- Root cause: **High Variance**
- Model overfit to quirks in training data
- Fix: add diverse fraud examples
- Or add regularization to constrain complexity

V28, page 15 — Bank B: diagnosing the problem

Question 3 of 3

A team can either simplify its overfitting model or collect a much larger, more varied dataset. Which does the video say is the better first move when it is possible?

- A** Simplify; data is expensive
- B** Collect the data; it reduces variance without adding bias
- C** Do both at once, always
- D** Neither; accept the errors

Hint. Which fix has no trade-off?

Answer your letter in the Zoom chat. We work through the answer together in the session.

FROM VIDEO V28

More Data Breaks the Tradeoff.

Adding diverse training examples reduces variance — without increasing bias.

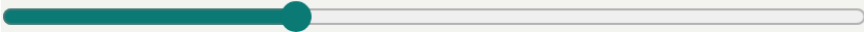


V28, page 9 — More data breaks the trade-off

Your turn: two errors, one dial, and data

Bias falls as the model gets more complex; variance rises, and rises faster when data is scarce. Total error is the sum. Move both sliders and watch where the sweet spot goes.

4 model complexity



1,000 rows of training data



17

total error at this setting (bias 9, variance 3, noise 5). With 1,000 rows the sweet spot is complexity 5.



Try: 100 rows, then 10,000 rows, at the same complexity. Then find the best complexity for each. More data moves the sweet spot to the right without touching bias.

Before we move on

Three things from this block

- 1 Bias is the wrong sight; variance is the unsteady hand.
- 2 Diagnose first: level of both scores, or the gap.
- 3 More diverse data helps without a trade. Everything else is a trade.

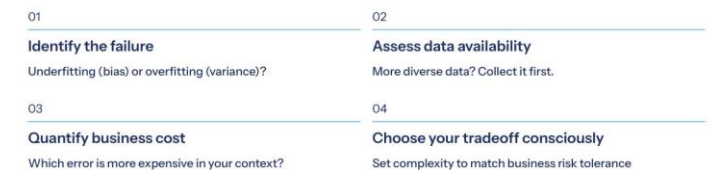
FROM VIDEO V28

Key Takeaways



V28, page 17 — Key takeaways

Your Decision Framework



V28, page 16 — Your decision framework

Block 13 of 13, 12 minutes. Video V29.

How complex should the model be?

Two questions decide it: how much data do you have, and what happens if the model is wrong?



the deck you watched: V29, Choosing model complexity

The one idea

Two questions decide it: how much data do you have, and what happens if the model is wrong?

Data size

Your complexity budget. Roughly 10 to 20 examples per adjustable part. 500 rows buys a small model.

Stakes

A wrong film suggestion costs nothing. A wrong loan or diagnosis costs someone their home or health.

Add regulation as a third: in lending, insurance and healthcare, a model that cannot explain itself may be illegal, however accurate.

FROM VIDEO V29

The Answer Depends on Exactly Two Things



How Much Data You Have

Data size sets your complexity budget.



What's at Stake If You're Wrong

Business risk determines how much variance you can tolerate.

V29, page 5 — The answer depends on exactly two things

Your Complexity Budget

The Rule

You need roughly 10-20 examples per parameter to estimate reliably.

Data size is the ceiling on model complexity.

500 rows

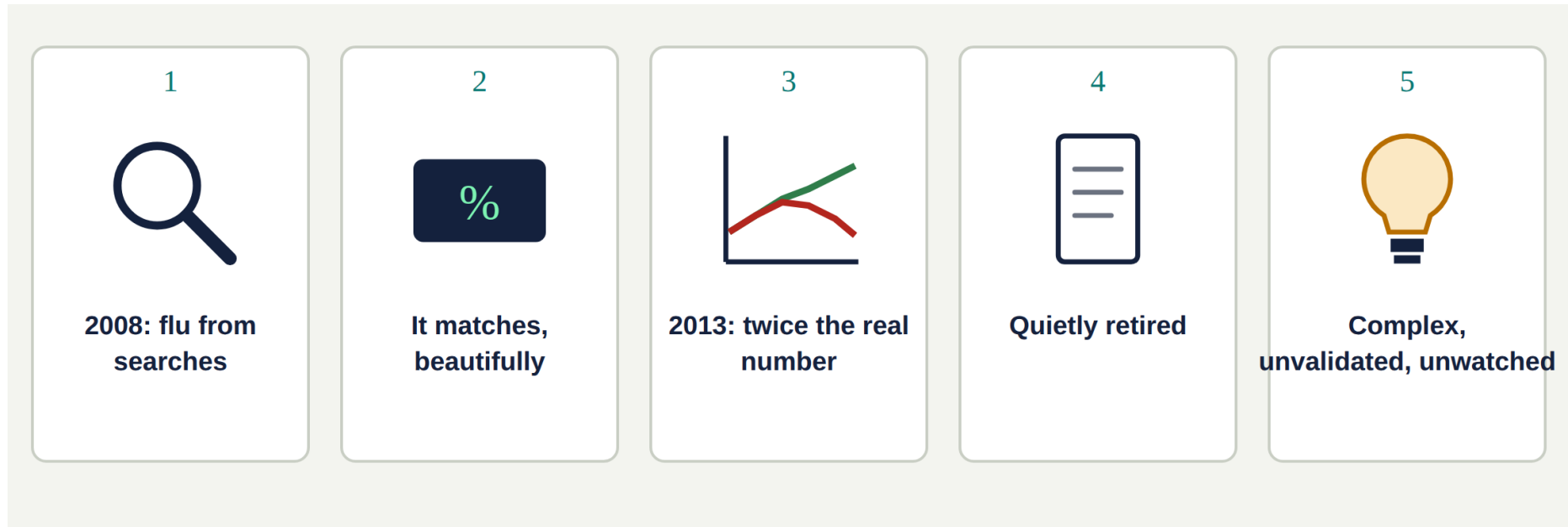
→ 25-50 parameters max

1M rows

→ Tens of thousands of parameters

V29, page 7 — Your complexity budget

A story: the flu model that was too clever



- 1 A large search company notices that flu-related searches rise when flu does. It builds a model on billions of searches to predict outbreaks weeks before the health authorities.
- 2 For a few seasons the model tracks the official flu numbers closely. Journals publish it. It is hailed as the future of public health.
- 3 Then it predicts almost double the flu that actually happens. Search habits had changed, media coverage had changed, the model had learned patterns that no longer held.
- 4 The service is shut down two years later. The lesson is written up in a scientific journal under the title "the parable" of the project.
- 5 Lesson: huge data and a complex model are not a substitute for honest validation, and no model survives a changing world without being watched.

If you remember nothing else

Three things to keep

- 1 Data size sets the ceiling on complexity: small data, simple model.
- 2 Stakes set the floor on explainability: high stakes, a model you can explain.
- 3 Training accuracy is a mirage. Validation on held-out data is the truth.

FROM VIDEO V29



The Central Lesson

The right model is never the one with the highest training accuracy. It's the one that **balances generalization error and business risk.**

Next Up

Measuring generalization — and spotting drift when the real world shifts beneath your model.

V29, page 20 — The central lesson

Training accuracy is a
mirage.

Validation error on held-out data is truth.

V29, page 13 — Training accuracy is a mirage

Question 1 of 3

A churn project has 500 rows of customer data. The team wants to build a ten-layer neural network because deep learning is in fashion. What does the complexity budget say?

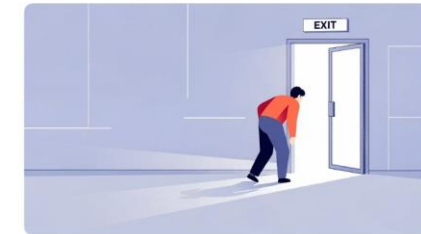
- A** Go ahead, more layers means more accuracy
- B** Use a simple model; 500 rows cannot support that many adjustable parts
- C** Collect 500 more rows, then build the network
- D** Use the network but lower the threshold

Hint. Examples per adjustable part.

Answer your letter in the Zoom chat. We work through the answer together in the session.

FROM VIDEO V29

CASE STUDY 1 Customer Churn — 500 Rows



The Temptation

10-layer neural network — deep learning is in vogue.

The Answer

Resist it. Logistic regression or a shallow decision tree wins here.

Why

Not enough signal for complexity to find real patterns.

V29, page 15 — Case study 1: customer churn, 500 rows

Question 2 of 3

Credit risk, 50,000 loans, regulated lender. A 99% black box or a 94% model whose decisions can be explained to a regulator?

- A** The 99% model; accuracy is what matters
- B** The 94% model; in lending, explaining a decision is a legal requirement
- C** The 99% model, with a disclaimer
- D** Flip a coin; they are close

Hint. Who can ask "why"?

Answer your letter in the Zoom chat. We work through the answer together in the session.

FROM VIDEO V29

CASE STUDY 2

Credit Risk — 50,000 Loans



High stakes. Real consequences.

Wrong decisions hurt real people.

The Answer

Gradient-boosted tree with regularization.

- Complex enough to beat linear models
- Interpretable enough for regulators

V29, page 16 — Case study 2: credit risk, 50,000 loans

Question 3 of 3

A recommendation engine has billions of interactions and a wrong suggestion costs almost nothing. Which complexity tier is justified?

- A** A simple model, to be safe
- B** Full complexity, because the data is huge and the errors are cheap
- C** Medium complexity, always the safe middle
- D** No model; use a rule of thumb

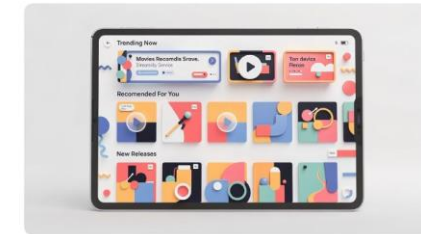
Hint. Big data, cheap mistakes.

Answer your letter in the Zoom chat. We work through the answer together in the session.

FROM VIDEO V29

CASE STUDY 3

Netflix Recommendations — Billions of Interactions



Cost of being wrong?

A suboptimal suggestion. Practically zero consequence.

The Answer

Deep neural network. The scale and forgivability of errors justify full complexity.

V29, page 17 — Case study 3: Netflix recommendations

Your turn: the decision matrix

Three questions about your project. The card answers with a complexity tier and the reason, the way the video's case studies do.

How much data?

Medium, 1,000 to 100,000 rows ▾

What if the model is wrong?

Serious: money, health, a home ▾

Regulated decision?

Yes (lending, insurance, health) ▾

Middle tier: boosted trees with constraints, with explanations for every decision

recommended tier

Enough data to beat a straight line, not enough to justify a black box. High stakes or regulation mean every refusal must be explainable.

Avoid: an unexplainable model, whatever its accuracy.

Try: the three cases from the video: 500 rows of churn, 50,000 regulated loans, billions of film ratings.

Before we move on

Three things from this block

- 1 Data size is the ceiling; stakes are the floor.
- 2 The right model is never the one with the best training score.
- 3 Watch it after launch: the world moves.

FROM VIDEO V29



The Central Lesson

The right model is never the one with the highest training accuracy.
It's the one that **balances generalization error and business risk.**

Next Up

Measuring generalization — and spotting drift when the real world shifts beneath your model.

V29, page 20 — The central lesson

The Complexity Decision Matrix

Data Size	Stakes	Model Choice
Small (<1K rows)	High	Logistic Regression
Medium (1K-100K)	Moderate	Gradient Boosting + Regularization
Large (>100K rows)	Low	Deep Neural Network

V29, page 19 — The complexity decision matrix

Before Session 3, Tuesday 15 September

Do

- Watch the Session 3 videos (V30 to V39).
- Open this file again and read the speaker notes under the slides where a term was new to you.
- Bring one example from your own work where the two kinds of mistake cost different amounts.

Assessed

- Quiz 2 (Modules 3 and 4): before 15 September.
- Quiz 3 (Modules 5 and 6): before 17 September.
- Midterm exam (Modules 1 to 6, 90 minutes, online): in the 18 to 23 September gap. It includes reading a confusion matrix and diagnosing a validation curve. No coding.

Questions. The interpretation runs a few seconds behind, so anything you ask out loud may arrive after we have moved on. Send questions in the Zoom chat during the session, or afterwards on the WhatsApp number you already have. Nothing is too small to ask.

Words used today, in plain language

Each video title is a link. All of them are short: watch the one whose word you are least sure about.

accuracy	the share of all predictions that were right; it hides which way the model was wrong
bias-variance trade-off	too rigid (bias) or too flexible (variance); every complexity choice moves one up and the other down
cherry-picking, or the multiple comparisons problem	trying many measures and reporting the flattering one; painting the target after shooting
classification	predicting a category: yes or no, fraud or honest
confusion matrix	four counts: what the model said, against what was true
cross-validation	repeating the hold-out test several times with different slices of the data

Words used today, in plain language (continued)

F1 score	one number that balances precision and recall
false alarm (false positive)	the model said yes, the truth was no
hold-out test set	data kept aside and never used for training, so the test is honest
imbalanced data	one answer is far more common than the other, so the common answer scores high by itself
learning curve	error plotted against the amount of training data; tells you whether more data would help
MAE, mean absolute error	the average size of the misses when predicting a number; every miss counts the same

Words used today, in plain language (continued)

miss (false negative)

the model said no, the truth was yes

overfitting

a model, or a team, that memorises the past and fails on anything new

precision

of everything the model flagged, the share that was right

recall

of all the real cases, the share the model caught

RMSE, root mean squared error

like MAE, but big misses count much more

success metric

the one number the business and the data team agree to judge the model by, fixed before anything is built

Words used today, in plain language (continued)

threshold

the cut-off score above which the model answers yes; moving it trades false alarms for misses

underfitting

a model too simple to see the pattern: mediocre on old data and new alike

validation curve

training and validation error plotted against model complexity; the bottom of the validation line is the setting to use